MoE-Loco: Mixture of Experts for Multitask Locomotion

Runhan Huang*[1,2], Shaoting Zhu*[1,2], Yilun Du[3], Hang Zhao†[1,2]
MoE-Loco.github.io/

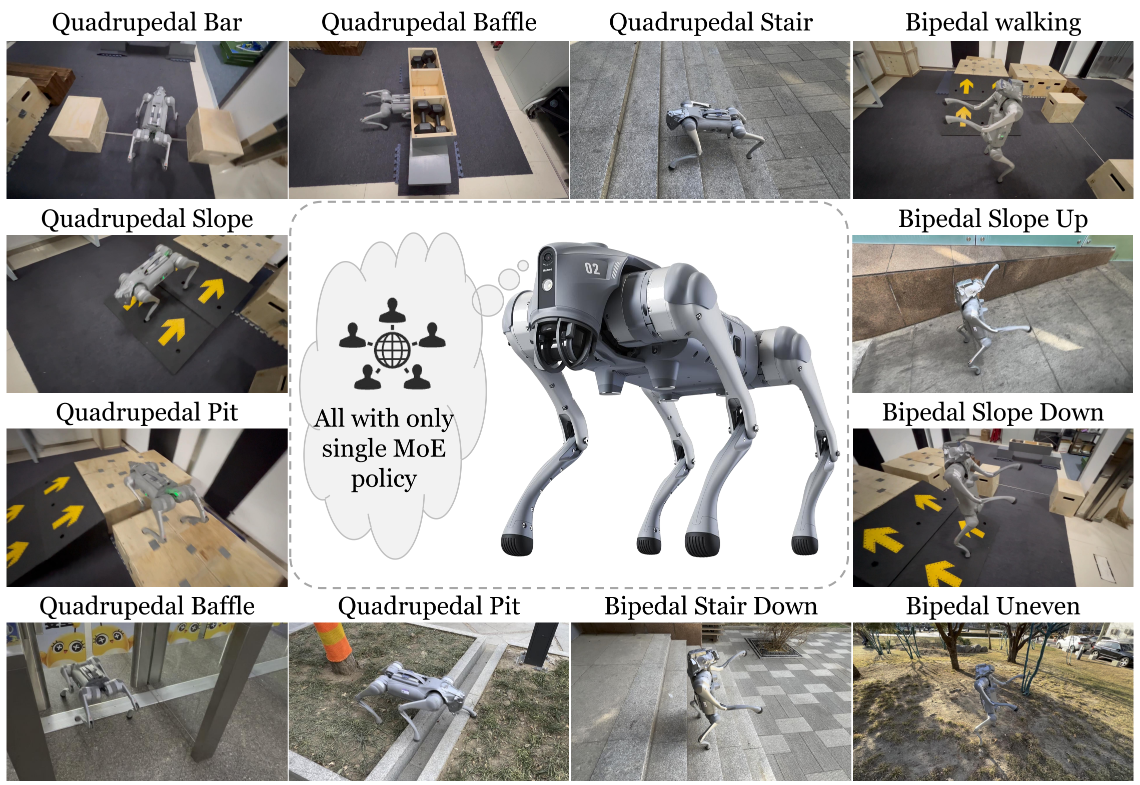

Fig. 1: **We introduce MoE-Loco.** With a single MoE policy, quadruped robot can traverse a variety of challenging terrains and perform different locomotion modes, including bipedal and quadrupedal gaits.

***Abstract*— We present MoE-Loco, a Mixture of Experts (MoE) framework for multitask locomotion for legged robots. Our method enables a single policy to handle diverse terrains, including bars, pits, stairs, slopes, and baffles, while supporting quadrupedal and bipedal gaits. Using MoE, we mitigate the gradient conflicts that typically arise in multitask reinforcement learning, improving both training efficiency and performance. Our experiments demonstrate that different experts naturally specialize in distinct locomotion behaviors, which can be leveraged for task migration and skill composition. We further validate our approach in both simulation and real-world deployment, showcasing its robustness and adaptability.**

I. INTRODUCTION

Robots are often required to traverse diverse terrains and demonstrate various skills [1], [2]. Recent advancements in reinforcement learning (RL) algorithms and physics-based simulators have enabled RL-based approaches to become the dominant paradigm for training robot locomotion policies [3]–[7]. However, while single-task RL has demonstrated remarkable success, learning a unified policy that generalizes across multiple tasks, terrains, and locomotion modes remains a significant challenge.

Recent research has explored training locomotion policies with diverse skills by having diverse terrains in simulation for parallel training [8]–[10]. However, multitask RL with a simple neural network architecture often suffers from gradient conflicts [11], [12], which leads to inferior model performance. What is worse, training a policy across multiple terrains with different gaits poses further challenges, leading to model divergence.

In this work, we enable a quadruped robot to traverse various terrains—including bars, pits, stairs, slopes, and baffles—while also supporting gait switching between bipedal and quadrupedal modes, using only one policy. We integrate the Mixture of Experts (MoE) framework [13]–[16] as a modular network structure for multitask locomotion reinforcement learning. We demonstrate that the MoE framework alleviates gradient conflicts by directing gradients to specialized experts, thus improving training efficiency and overall performance. Furthermore, we analyze the roles of

[1]IIIS, Tsinghua University, Beijing, China
[2]Shanghai Qi Zhi Institute, Shanghai, China
[3]Massachusetts Institute of Technology, MA, USA
* These authors contributed equally to this work.
† Corresponding at: hangzhao@mail.tsinghua.edu.cn

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MoE-Loco：用于多任务运动的专家混合

黄润涵*[1,2], 朱少廷*[1,2],杜逸伦[3],赵航†[1,2]MoE-Loco.github.io/

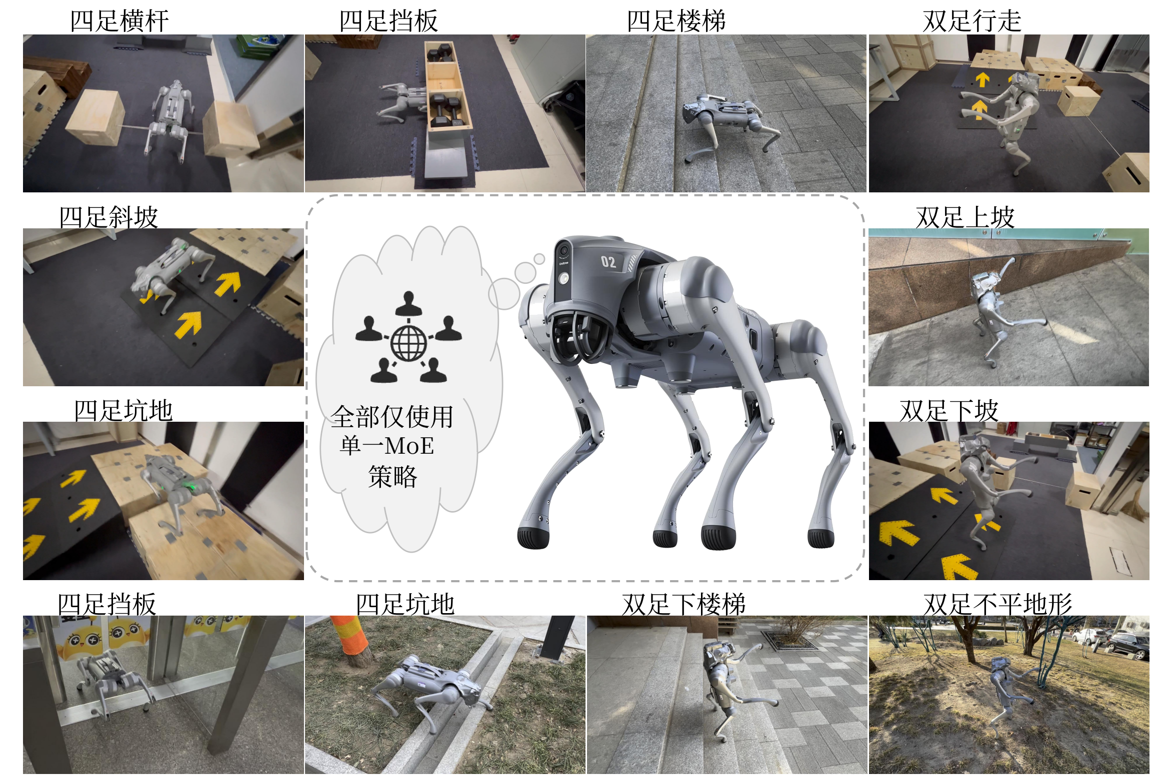

图1：**我们提出**MoE-Loco。 凭借单一的MoE策略，四足机器人能够穿越各种具有挑战性的地形，并执行不同的运动模式，包括双足行走和四足步态。

摘要——**我们提出了 MoE-Loco，一个用于腿式机器人多任务运动的专家混合（MoE）框架。我们的方法使单一策略能够应对多种地形，包括栏杆、坑、楼梯、坡道和挡板，同时支持四足行走和双足步态。通过使用MoE，我们缓解了多任务强化学习中常见的梯度冲突，提高了训练效率和性能。我们的实验表明，不同的专家自然地专精于不同的运动行为，这可用于任务迁移和技能组合。我们进一步在仿真和实际部署中验证了我们的方法，展示了其鲁棒性和适应性。**

I. 引言

机器人通常需要穿越多样化的地形并展示各种技能 [1], [2]。近年来，强化学习（RL）算法和基于物理的仿真器的进步，使得基于强化学习的方法成为训练机器人运动策略的主导范式 [3]–[7]。然而，尽管单任务强化学习已展现出显著的成功，但学习一个能够跨多任务、地形和运动模式泛化的统一策略仍然是一个重大挑战。

近期研究通过在仿真中设置多样地形进行并行训练，探索了训练具有多种技能的运动策略 [8]–[10]。然而，使用简单神经网络架构的多任务强化学习常常遭受梯度冲突 [11], [12], ，导致模型性能不佳。更糟糕的是，在具有不同步态的多个地形上训练策略会带来进一步的挑战，导致模型发散。

在这项工作中，我们使四足机器人能够穿越各种地形——包括栏杆、坑、楼梯、坡道和障碍物——同时仅使用一个策略即可支持双足行走与四足行走模式之间的步态切换。我们将专家混合（MoE）框架 [13]–[16] 作为模块化网络结构，用于多任务运动强化学习。我们证明，MoE框架通过将梯度引导至专业化专家来缓解梯度冲突，从而提高训练效率和整体性能。此外，我们分析了

[1]IIIS，清华大学，中国北京[2]上海期智研究院，中国上海[3]麻省理工学院，美国马萨诸塞州 * 这些作者对这项工作贡献相同。
† 通讯作者：hangzhao@mail.tsinghua.edu.cn

different experts within the MoE model and observe that they naturally specialize in distinct behaviors. Leveraging this property, we can manually adjust the ratios between different experts to compose new skills, underscoring the adaptability and reusability of our modular approach for novel tasks.

In summary, our contributions are as follows.

- We train and deploy a single neural network policy that enables a quadruped robot to **cross challenging terrains** and perform fundamentally different locomotion modes, including **bipedal and quadrupedal gaits**.
- We integrate the **MoE architecture** into locomotion policy training to **mitigate gradient conflicts**, improve training efficiency, and overall model performance.
- We conduct qualitative and quantitative analysis of MoE, uncovering expert specialization patterns. Using these insights, we explore the potential of MoE for **task migration** and **skill composition**.

II. RELATED WORKS

A. Reinforcement Learning for Robot Locomotion

Using reinforcement learning for robot locomotion control has become increasingly popular in recent years. It has demonstrated the ability to learn legged locomotion behaviors in both simulation [17], [18] and the real world [3], [19]. Not only can it traverse a variety of complex terrains [8], [20], [21], but it can also achieve high-speed running [22], [23]. Furthermore, reinforcement learning has enabled robots to perform extreme tasks and master skills such as bipedal walking [24], [25], opening doors [1], [4], navigating rocky terrains [26], and even executing high-speed parkour in challenging environments [7], [9], [10], [27]. However, most of these works focus on specific skills and a limited number of terrains, rarely considering multitask learning.

B. MultiTask Learning

Multitask learning (MTL) aims to train a unified network that can perform across different tasks [28]–[30]. MTL allows multiple tasks to benefit from shared knowledge [31], [32], but some works highlight the challenge of negative gradient conflicts during training [33]–[35]. Multitask Reinforcement Learning (MTRL) is one of the most popular areas of research in this domain. Many algorithms have been developed to improve the effectiveness of MTRL [36]–[38]. Moreover, MTRL is widely applied in the robotics field, though much of the focus has been on manipulation tasks [16], [35], [39], [40]. In the context of locomotion, works like ManyQuadrupeds [41] focus on learning a unified policy for different categories of quadruped robots. MELA [42] employs pretrained expert models to construct a locomotion policy, although it primarily concentrates on basic skill acquisition. Moreover, the pretraining process for specialized neural network policies requires substantial reward engineering efforts. MTAC [43] attempts to train a policy across different terrains using hierarchical RL, but their approach can only handle one gait with three relatively simple terrains and has not been deployed on a real robot.

C. Mixture of Experts (MoE)

The concept of Mixture of Experts (MoE), originally introduced in [13], [44], has received extensive attention in recent years [45], [46]. It has found widespread application in fields such as natural language processing [47], [48], computer vision [49], [50], and multi-modal learning [51], [52]. MoE has also been applied in reinforcement learning and robotics. DeepMind [14] has explored using MoE to scale reinforcement learning. MELA [39] proposed a Multi-Expert Learning Architecture to generate adaptive skills from a set of representative expert skills, but their focus is primarily on simple actions.

III. METHOD

A. Task Definition

In this paper, we focus on 9 challenging locomotion tasks, encompassing both quadrupedal and bipedal gaits. The quadrupedal gait tasks include bar crossing, pit crossing, baffle crawling, stair climbing, and slope walking. The bipedal gait tasks consist of standing up, plane walking, slope walking, and stair descending. Our terrains in the simulation environment are shown in Figure 2. The robot is controlled via velocity commands from a joystick, where a one-hot vector is used to indicate whether to walk in the quadrupedal or bipedal gait.

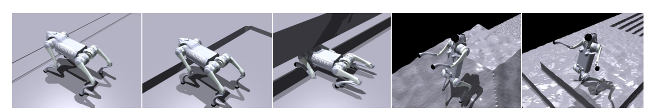

Fig. 2: **A snapshot of the terrain settings.** From left to right: bar, pit, baffle, slope, and stairs.

We define multitask locomotion as a Markov Decision Process (MDP), represented by the tuple $\langle S_\tau, A_\tau, T_\tau, R_\tau, \gamma_\tau \rangle$. In our training framework, the multitask nature is characterized by several key aspects. First, different locomotion terrains correspond to distinct subsets of the state space, denoted as S_τ, where $S_\tau \subseteq S$ represents the states relevant to a specific task. However, the full state space S remains unknown to the robot. For instance, the task of walking on slopes involves a state space that differs from those required for stair climbing or bar traversal.

Moreover, the reward function R varies across different gaits, reflecting task-specific objectives. Additionally, the termination conditions depend on the type of gait, leading to distinct transition dynamics T. The robot learns a policy $\pi(a|s)$ that selects actions based on both the terrain and gait, aiming to maximize the cumulative reward across tasks:

$$J(\pi) = \mathbb{E}\left[\sum_{\tau}\sum_{t=0}^{\infty} \gamma^t R_\tau(s_t, a_t, s_{t+1})\right] \quad (1)$$

The goal is to learn a single universal policy that generalizes across various tasks. We demonstrate our method in the condition of blind locomotion (only use proprioception as input). In fact, our approach can also be incorporated into visual RL locomotion settings, further enhancing their multitask locomotion capabilities.

MoE模型中不同专家的作用，并观察到它们自然地专精于不同的行为。利用这一特性，我们可以手动调整不同专家之间的比例来组合新技能，这凸显了我们模块化方法在新任务中的适应性和可重用性。

总之，我们的贡献如下。

- 我们训练并部署了一个单一的神经网络策略，使四足机器人能够**穿越具有挑战性的地形**，并执行根本不同的运动模式，包括**双足行走和四足步态**。
- 我们将**MoE架构**集成到运动策略训练中，以**缓解梯度冲突**，提高训练效率和整体模型性能。
- 我们对MoE进行了定性和定量分析，揭示了专家特化模式。利用这些见解，我们探索了MoE在**任务迁移**和**技能组合**方面的潜力。

II. 相关工作

A. 机器人运动的强化学习

近年来，利用强化学习进行机器人运动控制已变得越来越流行。它已展现出在仿真 [17], [18] 和真实世界 [3], [19]中学习腿部运动行为的能力。它不仅能够穿越各种复杂地形 [8], [20], [21], ，还能实现高速奔跑 [22], [23]。此外，强化学习使机器人能够执行极限任务并掌握诸如双足行走 [24], [25],、开门 [1], [4],、在崎岖地形中导航 [26], ，甚至在具有挑战性的环境中执行高速跑酷 [7], [9], [10], [27]等技能。然而，这些工作大多聚焦于特定技能和有限的地形，很少考虑多任务学习。

B. 多任务学习

多任务学习（MTL）旨在训练一个统一的网络，使其能够跨不同任务执行 [28]–[30]。MTL允许多个任务从共享知识中受益 [31], [32],，但一些研究也强调了训练过程中负梯度冲突的挑战 [33]–[35]。多任务强化学习（MTRL）是该领域最热门的研究方向之一。许多算法已被开发出来以提高MTRL的有效性 [36]–[38]。此外，MTRL在机器人领域得到了广泛应用，尽管大部分研究重点集中在操作任务上 [16], [35], [39], [40]。在运动控制方面，像ManyQuadrupeds [41]这样的工作专注于为不同类别的四足机器人学习统一策略。MELA [42]采用预训练的专家模型来构建运动策略，尽管它主要关注基础技能获取。此外，针对专门神经网络策略的预训练过程需要大量的奖励工程工作。MTAC [43] 尝试使用分层强化学习在不同地形上训练策略，但他们的方法只能处理一种步态和三种相对简单的地形，并且尚未在真实机器人上部署。

C. 专家混合（*MoE*）

专家混合（MoE）的概念最初在 [13], [44],中提出，近年来受到了广泛关注 [45], [46]。它已在自然语言处理 [47], [48],、计算机视觉 [49], [50],和多模态学习 [51], [52]等领域得到广泛应用。MoE也被应用于强化学习和机器人学。DeepMind [14] 探索了使用MoE来扩展强化学习。MELA [39] 提出了一种多专家学习架构，从一组代表性专家技能中生成自适应技能，但他们的研究重点主要在于简单动作。

III. 方法

A. 任务定义

在本文中，我们聚焦于9项具有挑战性的运动任务，涵盖四足行走和双足步态。四足步态任务包括跨栏、跨坑、挡板爬行、爬楼梯和坡道行走。双足步态任务包括站立、平地行走、坡道行走和下楼梯。我们在仿真环境中的地形如图2所示。机器人通过操纵杆发出的速度指令进行控制，其中使用独热向量来指示是以四足步态还是双足步态行走。

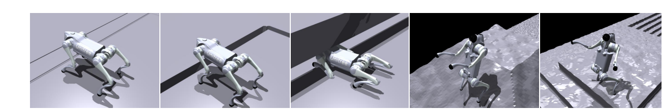

图2：**地形设置快照**。从左到右依次为：栏杆、坑、挡板、坡道和楼梯。

我们将多任务运动定义为马尔可夫决策过程（MDP），用元组$S_\tau, A_\tau, T_\tau, R_\tau, \gamma_\tau$表示。在我们的训练框架中，多任务特性通过几个关键方面体现。首先，不同的运动地形对应状态空间的不同子集，记为S_τ，其中 $S_\tau \subseteq S$ 表示与特定任务相关的状态。然而，完整的状态空间 S 对机器人而言是未知的。例如，在坡道上行走的任务涉及的状态空间与爬楼梯或跨越栏杆所需的状态空间不同。

此外，奖励函数 R 在不同步态间有所变化，反映了任务特定的目标。同时，终止条件取决于步态类型，导致不同的转移动力学 T。机器人学习策略$\pi(a|s)$，该策略根据地形和步态选择动作，旨在最大化跨任务的累积奖励：

$$J(\pi) = \mathbb{E}\left[\sum_{\tau}\sum_{t=0}^{\infty} \gamma^t R_\tau(s_t, a_t, s_{t+1})\right] \quad (1)$$

目标是学习一个能够泛化到各种任务的单一通用策略。我们在盲态运动条件下（仅使用本体感觉作为输入）展示了我们的方法。实际上，我们的方法也可以融入视觉强化学习运动设置中，进一步增强其多任务运动能力。

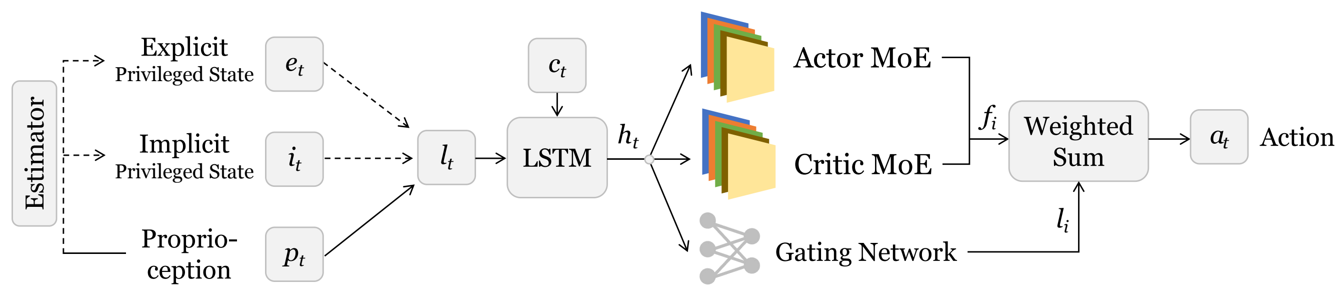

Fig. 3: **Overview of our MoELoco pipeline.** With the design of MoE architecture, our policy achieves robust multitask locomotion ability on various challenging terrains with multiple gaits.

B. MoE Based Multitask Locomotion Learning

In this section, we introduce our MoE based multitask locomotion learning. We incorporate a two-stage training framework following [20], and use PPO [53] as our reinforcement learning algorithm.

State Space: The entire process includes four types of observations: proprioception $\boldsymbol{p}_t$, explicit privileged state $\boldsymbol{e}_t$, implicit privileged state $\boldsymbol{i}_t$, and command $\boldsymbol{c}_t$. **1) Proprioception** $\boldsymbol{p}_t$ includes projected gravity and base angular velocity from the IMU, joint positions, joint velocities, and the last action. **2) Explicit privileged state** $\boldsymbol{e}_t$ contains the base linear velocity (IMU data is too noisy to use) and ground friction. **3) Implicit privileged state** $\boldsymbol{i}_t$ includes contact force of different robot link, which must be encoded into a low-dimensional latent representation to mitigate the sim-to-real gap [20]. **4) Command** $\boldsymbol{c}_t$ consists of a velocity command $V = (v_x, v_y, v_{\text{yaw}})$ and a one-hot vector g, where $g = 0$ represents a quadrupedal gait and $g = 1$ represents a bipedal gait in the context of multitask reinforcement learning.

Action Space: The action space $\boldsymbol{a}_t \in \mathbb{R}^{12}$ consists of the desired joint positions for all 12 joints.

Reward Design: Under the setting of the multitask learning, the robot receives different rewards based on the gait command g. For quadrupedal locomotion ($g = 0$), the total reward is defined as $r^{\text{quad}} = r^{\text{quad}}_{\text{track}} + r^{\text{quad}}_{\text{reg}}$. For bipedal locomotion ($g = 1$), the total reward is given by $r^{\text{bip}} = r^{\text{bip}}_{\text{track}} + r^{\text{bip}}_{\text{stand}} + r^{\text{bip}}_{\text{reg}}$. The detailed reward functions can be found in Appendix V-A.

Termination: The robot terminates under different circumstances under different gait modes. When $g = 0$, the robot terminates when $\theta_{\text{roll}} > 1.0$ or $\theta_{\text{pitch}} > 1.6$. When $g = 1$, the robot terminates when any other links except rear feet and calf contacts the ground after 1 second.

Training: Our training primarily follows the Probability Annealing Selection (PAS) paradigm [54]. Overall, the robot utilizes both privileged and proprioceptive information in the first stage; but in the second stage, it learns to rely exclusively on proprioception for locomotion, using an estimator to estimate the privileged latent. In the first training stage, all observation states $[\boldsymbol{p}_t, \boldsymbol{e}_t, \boldsymbol{i}_t, \boldsymbol{c}_t]$ are accessible to train an Oracle policy. The implicit state $\boldsymbol{i}_t$ is first encoded through an encoder network into a latent representation, which is then concatenated with the explicit privileged state $\boldsymbol{e}_t$ and proprioception $\boldsymbol{p}_t$ to form the dual-state representation $\boldsymbol{l}_t = [\text{Enc}(\boldsymbol{i}_t), \boldsymbol{e}_t, \boldsymbol{p}_t]$. Then, the downstream LSTM integrates historical information into state $\boldsymbol{h}_t$. As discussed in subsection IV-C, jointly learning multiple tasks in multitask reinforcement learning (MTRL) often leads to gradient conflicts. To address this issue, we incorporate the Mixture of Experts (MoE) architecture into both the actor and critic networks, effectively mitigating gradient conflicts and improving learning efficiency. Specifically, each MoE module f operates as follows:

$$\hat{g}_i = \text{softmax}(g(\boldsymbol{h}_t))[i], \tag{2}$$

$$\boldsymbol{a}_t = \sum_{i=1}^{N} \hat{g}_i \cdot f_i(\boldsymbol{h}_t), \tag{3}$$

Here, $\boldsymbol{h}_t$ is the output of the low-level LSTM module, g is the gating network that outputs the gating scores, and f_i denotes expert i. Additionally, we pretrain the estimator module in this stage using a L2 loss L_{recon} to reconstruct Estimator$(\boldsymbol{p}_t, \boldsymbol{c}_t)$ into $[\text{Enc}(\boldsymbol{i}_t), \boldsymbol{e}_t]$. In summary, the overall optimization objective is:

$$L_{\text{surro}} + L_{\text{value}} + L_{\text{recon}}, \tag{4}$$

where L_{surro} and L_{value} are surrogate loss and value loss in PPO algorithm.

In the second training stage, the policy can only access $[\boldsymbol{p}_t, \boldsymbol{c}_t]$ as observations. The weights of the estimator, the low-level LSTM, and the MoE modules are initialized by copying them from the first training stage. Probability Annealing Selection [54] is then employed to gradually adapt the policy to inaccurate estimates with minimal degradation of the Oracle policy performance. Detailed pseudocode is in subsection V-B

The MoE architecture facilitates the coordination of similar task skills while minimizing conflicts between heterogeneous tasks by dynamically routing tasks to appropriate experts. This automatic routing enables specialization, improving both efficiency and task performance. Additionally, we incorporate MoE into the critic network to better capture diverse task reward structures. The actor MoE and critic MoE shares the same gating network. By using a shared gating network, we ensure consistency between policy evaluation and action generation.

C. Skill Decomposition and Composition

A key challenge in multitask learning is how to efficiently utilize previous acquired locomotion skills to form new locomotion tasks. The MoE framework offers a natural solution by dynamically decompose tasks into expert of different skills. Specifically, we conduct quantitative analyses on expert

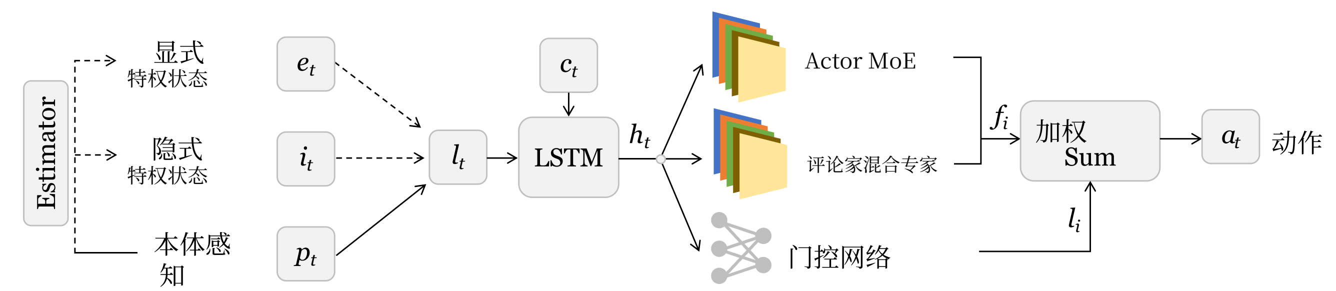

图3：**我们MoELoco 流程的概览**。通过MoE架构的设计，我们的策略在多种具有挑战性的地形上实现了稳健的多任务运动能力，支持多种步态。

B. 基于*MoE*的多任务运动学习

In 基于多任务运动学习。我们采用了一个两阶段训练框架，遵循 [20],，并使用PPO [53] 作为我们的强化学习算法。

状态空间： 整个过程包括四种类型的观测：本体感觉 $\boldsymbol{p}_t$、显式特权状态 $\boldsymbol{e}_t$、隐式特权状态 $\boldsymbol{i}_t$和指令 $\boldsymbol{c}_t$。**1) 本体感觉** $\boldsymbol{p}_t$ 包括来自惯性测量单元的投影重力和基座角速度、关节位置、关节速度以及上一次动作。**2) 显式特权状态** $\boldsymbol{e}_t$ 包含基座线速度（IMU数据噪声过大，无法使用）和地面摩擦力。**3) 隐式特权状态** $\boldsymbol{i}_t$ 包括不同机器人连杆的接触力，必须编码为低维潜在表示，以缓解仿真到现实差距 [20]。**4) 指令** $\boldsymbol{c}_t$ 由速度指令 $V =$（v_x, v_y,v_{yaw}）和独热向量 g组成，其中 $g = 0$ 代表四足步态，$g = 1$ 代表双足步态，在多任务强化学习的背景下。

动作空间： 动作空间 $\boldsymbol{a}_t \in \mathbb{R}^{12}$ 由所有12个关节的期望关节位置组成。

奖励设计： 在多任务学习的设定下，机器人根据步态指令接收不同的奖励g。对于四足运动（$g = 0$），总奖励定义为 $r^{\text{quad}} = r^{\text{quad}}$ 跟踪+ $r^{\text{quad}}_{\text{reg}}$。对于双足运动（ $g =$ 1 $r^{\text{bip}} = r^{\text{bip}} + r^{\text{bip}}$ ），总奖励由跟踪站立+ $r^{\text{bip}}_{\text{reg}}$给出。详细的奖励函数可在附录V-A中找到。

终止： 在不同的步态模式下，机器人会在不同的情况下终止。当 $g = 0$时，机器人在 $\theta_{\text{roll}} > 1$。0或 $\theta_{\text{pitch}} > 1$。6时终止。当 $g = 1$时，机器人在1秒后除后脚和小腿外的任何其他连杆接触地面时终止。

训练： 我们的训练主要遵循概率退火选择（PAS）范式 [54]。总体而言，在第一阶段，机器人同时利用特权状态和本体感觉信息；但在第二阶段，它学习仅依赖本体感觉进行运动，使用估计器来估计特权潜在变量。在第一训练阶段，所有观测状态 $[\boldsymbol{p}_t, \boldsymbol{e}_t, \boldsymbol{i}_t, \boldsymbol{c}_t]$均可用于训练Oracle策略。隐式状态 $\boldsymbol{i}_t$ 首先通过编码器网络编码为潜在表示，然后与显式特权状态 $\boldsymbol{e}_t$ 和本体感觉信息$\boldsymbol{p}_t$ 拼接，形成双状态表示 $\boldsymbol{l}_t = [\text{Enc}(\boldsymbol{i}_t)$，$\boldsymbol{e}_t$，$\boldsymbol{p}_t]$。随后，下游LSTM将历史信息整合到状态 $\boldsymbol{h}_t$中。如第IV-C小节所述，联合学习多任务强化学习（MTRL）中的多任务往往会导致梯度冲突。为解决这一问题，我们在演员网络和评论家网络中均引入了专家混合（MoE）架构，有效缓解了梯度冲突并提高了学习效率。具体而言，每个MoE模块 f 的运行方式如下：

$$\hat{g}_i = \text{softmax}(g(\boldsymbol{h}_t))[i], \tag{2}$$

$$\boldsymbol{a}_t = \sum_{i=1}^{N} \hat{g}_i \cdot f_i(\boldsymbol{h}_t), \tag{3}$$

这里，$\boldsymbol{h}_t$ 是低级LSTM模块的输出，g 是输出门控分数的门控网络，f_i 表示专家 i。此外，我们在此阶段使用L2损失 L_{recon} 预训练估计器模块，以将估计器（$\boldsymbol{p}_t$, $\boldsymbol{c}_t$）重建为 $[\text{Enc}(\boldsymbol{i}_t), \boldsymbol{e}_t]$。总之，整体优化目标为：

$$L_{\text{surro}} + L_{\text{value}} + L_{\text{recon}}, \tag{4}$$

其中 L_{surro} 和 L_{value} 分别是PPO算法中的替代损失和价值损失。

在第二个训练阶段，策略只能访问$[\boldsymbol{p}_t, \boldsymbol{c}_t]$作为观测。估计器、低层LSTM和MoE模块的权重通过从第一个训练阶段复制来初始化。随后采用概率退火选择 [54]，以最小化Oracle策略性能的退化，逐步使策略适应不准确的估计。详细伪代码见第V-B小节

MoE架构通过动态地将任务路由到合适的专家，促进了相似任务技能的协调，同时最小化了异构任务之间的冲突。这种自动路由实现了专业化，提高了效率和任务性能。此外，我们将MoE融入评论家网络，以更好地捕捉多样化的任务奖励结构。Actor MoE和评论家混合专家共享同一个门控网络。通过使用共享门控网络，我们确保了策略评估和动作生成之间的一致性。

C. 技能分解与组合

多任务学习中的一个关键挑战是如何高效利用先前习得的运动技能来形成新的运动任务。MoE框架通过将任务动态分解为不同技能的专家，提供了一种自然的解决方案。具体而言，我们对专家

TABLE I: **Quantitative Comparison in Simulation.** Metrics include success rate, average travel distance, and average passing time.

Method	Mix	Bar (q)	Baffle (q)	Stair (q)	Pit (q)	Slope (q)	Walk (b)	Slope (b)	Stair (b)
	Success Rate ↑								
Ours	**0.879**	**0.886**	**0.924**	**0.684**	**0.902**	0.956	**0.932**	**0.961**	**0.964**
Ours w/o MoE	0.571	0.848	0.264	0.568	0.698	**0.988**	0.826	0.504	0.453
RMA	0.000	0.871	0.058	0.017	0.017	0.437	0.000	0.000	0.000
	Average Pass Time (s) ↓								
Ours	**230.98**	**102.42**	**87.84**	**179.14**	**91.86**	76.75	**92.37**	**86.14**	**86.44**
Ours w/o MoE	315.47	125.46	318.68	214.52	161.38	**65.28**	156.76	236.67	253.62
RMA	400.00	107.84	385.25	395.34	394.49	272.06	400.00	400.00	400.00
	Average Travel Distance (m) ↑								
Ours	**89.41**	**28.05**	**28.02**	20.42	**27.82**	27.62	**27.20**	**27.99**	**28.04**
Ours w/o MoE	57.12	27.59	17.41	**22.66**	25.59	**28.49**	22.73	26.21	14.23
RMA	13.40	27.39	11.31	3.92	12.48	21.33	2.00	2.00	2.00

coordination across various tasks. Detailed experiments and results are presented in subsection IV-E.

With the automatic decomposition of expert skills, we can recombine them with adjustable weights to synthesize new skills and gaits. Formally, we leverage pretrained experts and modify the gating weights as follows:

$$\hat{\boldsymbol{g}}_i = w[i] \cdot \mathrm{softmax}(g(\boldsymbol{h}_t))[i], \quad (5)$$

where $w[i]$ can be manually defined or dynamically adjusted by a neural network. This formulation enables controlled skill blending, allowing the robot to adapt and generalize to novel locomotion patterns.

Each expert naturally specializes in different aspects of movement, such as balancing, crawling, or obstacle crossing. By selectively adjusting the contributions of these experts, we can construct new locomotion strategies without requiring additional training. This recomposition process highlights the interpretability of MoE-based policies, as each expert's role can be explicitly identified and manipulated. Detailed experiments are provided in subsection IV-F and subsection IV-G.

IV. EXPERIMENTS

A. Experiment Setup

We conduct our simulation training in IsaacGym [17], utilizing 4096 robots concurrently on an NVIDIA RTX 3090 GPU. Training begins with 40,000 iterations for plane walking in both gaits, followed by 80,000 iterations on challenging terrain tasks. Finally, we apply PAS for 10,000 iterations to adapt the policy to pure proprioception input. The control frequency in both the simulation environment and the real world is 50Hz. Our policy is deployed on the Unitree Go2 quadruped robot, with an NVIDIA Jetson Orin serving as the onboard computing device. We use PD control for low-level joint execution ($K_p = 40.0, K_d = 0.5$). We select expert number N_{exp} as 6. In all experiment results, **q** represents **quadrupedal gait**, and **b** represents **bipedal gait**.

B. Multitask Performance

1) Simulation Experiment: We conduct comparative simulation experiments with the following baselines:

- **Ours w/o MoE** [20]: Uses the same framework as ours but replaces the MoE module with a simple MLP. We control the total parameter to be the same as our MoE policy.
- **RMA** [3]: Employs a 1D-CNN as an asynchronous adaptation module within a teacher-student training framework, without using an MoE module.

We constructed a benchmark for quadrupedal robot locomotion across different tasks. Our benchmark consists of a $5m \times 100m$ runway with various obstacles evenly distributed along the path. The obstacles include:

TABLE II: Benchmark tasks for simulation experiments

Obstacle Type	Specification	Gait Mode
Bars	5 bars, height: 0.05m – 0.2m	Quadrupedal
Pits	5 pits, width: 0.05m – 0.2m	Quadrupedal
Baffles	5 baffles, height: 0.3m – 0.22m	Quadrupedal
Up Stairs	3 sets, step height: 5cm – 15cm	Quadrupedal
Down Stairs	3 sets, step height: 5cm – 15cm	Quadrupedal
Up Slopes	3 sets, incline: 10° – 35°	Quadrupedal
Down Slopes	3 sets, incline: 10° – 35°	Quadrupedal
Plane	10m flat surface	Bipedal
Up Slopes	3 sets, incline: 10° – 35°	Bipedal
Down Slopes	3 sets, incline: 10° – 35°	Bipedal
Down Stairs	3 sets, step height: 5cm – 15cm	Bipedal

We also conducted experiments for each challenging tasks, each track is 30 meters long. We consider three metrics: **Success Rate**, **Average Pass Time**, and **Average Travel Distance**. A trial is considered successful if the robot reaches within 1m of the target point within 400 seconds. Upon completion, we record the travel time. Failure cases include falling off the runway, getting stuck, or meeting the termination conditions outlined in subsection III-B. For failed trials, the pass time is recorded as 400 seconds. We compute the overall success rate and average pass time across all trials. Additionally, we measure the average lateral travel distance of all robots at the end of the evaluation.

As shown in Table I, our MoE policy achieves the best performance in the mixed-task benchmark across all three metrics. In all single-task evaluations, except for quadrupedal slope walking, our policy outperforms others. This exception may be attributed to the relative simplicity of quadrupedal

表I：**仿真中的定量比较**。指标包括成功率、平均行驶距离和平均通过时间。

方法	Mix	栏杆 (q)	挡板 (q)	楼梯 (q)	坑 (q)	坡道 (q)	行走 (b)	坡道 (b)	楼梯 (b)
	成功率 ↑								
Ours	**0.879**	**0.886**	**0.924**	**0.684**	**0.902**	0.956	**0.932**	**0.961**	**0.964**
我们的方法（无MoE）	0.571	0.848	0.264	0.568	0.698	**0.988**	0.826	0.504	0.453
RMA	0.000	0.871	0.058	0.017	0.017	0.437	0.000	0.000	0.000
	平均通过时间（秒）↓								
Ours	**230.98**	**102.42**	**87.84**	**179.14**	**91.86**	76.75	**92.37**	**86.14**	**86.44**
我们的方法（无MoE）	315.47	125.46	318.68	214.52	161.38	**65.28**	156.76	236.67	253.62
RMA	400.00	107.84	385.25	395.34	394.49	272.06	400.00	400.00	400.00
	平均行驶距离（米）↑								
Ours	**89.41**	**28.05**	**28.02**	20.42	**27.82**	27.62	**27.20**	**27.99**	**28.04**
我们的方法（无MoE）	57.12	27.59	17.41	**22.66**	25.59	**28.49**	22.73	26.21	14.23
RMA	13.40	27.39	11.31	3.92	12.48	21.33	2.00	2.00	2.00

在各种任务中的协调进行了定量分析。详细的实验和结果见第IV-E小节。

通过专家技能的自动分解，我们可以使用可调权重将其重新组合，以合成新技能和步态。形式上，我们利用预训练专家并修改门控权重如下：

$$\hat{\boldsymbol{g}}_i = w[i] \cdot \mathrm{softmax}(g(\boldsymbol{h}_t))[i], \quad (5)$$

其中 $w[i]$ 可以手动定义或由神经网络动态调整。这种公式化方法实现了受控技能混合，使机器人能够适应并泛化到新型运动模式。

每个专家自然地在运动的不同方面表现出专长，例如平衡、爬行或越障。通过选择性地调整这些专家的贡献，我们可以在无需额外训练的情况下构建新的运动策略。这一重组过程凸显了基于MoE的策略的可解释性，因为每个专家的角色可以被明确地识别和操控。详细实验见第IV-F节和第IV-G节。

IV. 实验

A. 实验设置

我们在IsaacGym [17],中进行仿真训练，在NVIDIA RTX 3090 GPU上同时使用4096个机器人。训练首先进行40,000次迭代，用于两种步态的平地行走，随后在挑战性地形任务上进行80,000次迭代。最后，我们应用PAS进行10,000次迭代，以使策略适应纯本体感觉输入。仿真环境和真实世界中的控制频率均为50Hz。我们的策略部署在宇树Go2四足机器人上，以NVIDIA Jetson Orin作为机载计算设备。我们使用PD控制进行底层关节执行（$K_p = 40.0, K_d = 0.5$）。我们选择专家数量N_{exp}为6。在所有实验结果中，**q** 代表**四足步态**，**b** 代表**双足步态**。

B. 多任务性能

1) 仿真实验：我们与以下基线进行了对比仿真实验：

- **我们的方法（无MoE）** [20]：使用与我们的方法相同的框架，但将MoE模块替换为简单的MLP。我们控制总参数量与我们的MoE策略相同。
- **RMA** [3]：在教师-学生训练框架中采用1D-CNN作为异步自适应模块，不使用MoE模块。

我们构建了一个用于四足机器人跨不同任务运动的基准。我们的基准由一条$5m \times 100m$跑道组成，跑道上均匀分布着各种障碍物。障碍物包括：

表II：仿真实验的基准任务

障碍物类型	规格	步态模式
Bars	5 个栏杆，高度：0.05 米 – 0.2 米	四足行走
Pits	5 个坑，宽度：0.05 米 – 0.2 米	四足行走
挡板	5块挡板，高度：0.3米 – 0.22米	四足行走
上楼梯	3组，台阶高度：5厘米 – 15厘米	四足行走
下楼梯	3组，台阶高度：5厘米–15厘米	四足行走
上坡	3组，坡度：10° – 35°	四足行走
下坡	3组，坡度：10° – 35°	四足行走
平地	10米平坦表面	双足行走
上坡	3组，坡度：10° – 35°	双足行走
下坡	3组，坡度：10° – 35°	双足行走
下楼梯	3组，台阶高度：5厘米 – 15厘米	双足行走

我们还针对每项具有挑战性的任务进行了实验，每条跑道长30米。我们考虑三个指标：**成功率**，**平均通过时间**，以及**平均行驶距离**。如果机器人在400秒内到达目标点1米范围内，则该试验被视为成功。完成后，我们记录行驶时间。失败情况包括从跑道上跌落、卡住或满足第III-B小节中列出的终止条件。对于失败的试验，通过时间记录为400秒。我们计算所有试验的总体成功率和平均通过时间。此外，我们测量所有机器人在评估结束时的平均横向行驶距离。

如表I所示，我们的MoE策略在混合任务基准的三项指标中均取得了最佳表现。在所有单任务评估中，除四足斜坡行走外，我们的策略均优于其他方案。这一例外可能归因于四足

slope walking. Furthermore, policy without MoE struggles to effectively traverse the challenging multitask terrain setting, as it is significantly affected by gradient conflicts, as discussed in subsection IV-C. Regarding RMA, we adhere to its original implementation, which utilizes only an MLP backbone and a CNN encoder. This design choice leads to its suboptimal performance on multiple challenging terrains.

2) Real World Experiments: We deploy our MoE policy zero-shotly on real robots and conduct real world experiments. We test mix terrain that contains all challenging tasks, as well as each separated terrains. For the mix terrain, our robot first need to consequently cross 20cm bar, 22cm baffle, 15cm stairs, 20cm pits and 30 degree slopes in a quadrupedal gait. Then it receives a bipedal command to stand up, walk up the 30 degree slope, turn around and walk down. For each tests, we test for 20 trails and record the average success rate. We also test different policies for single tasks.

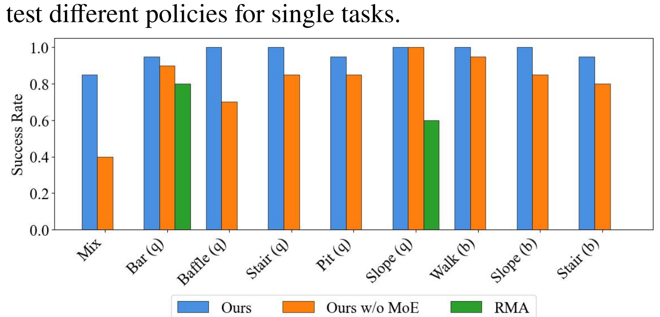

Fig. 4: **Real world success rate over multiple terrains and gaits.**

As shown in Figure 4, our method achieves better performance across all types of tasks and demonstrates a significantly higher success rate in mix terrain. Additionally, we conduct experiments in more outdoor environments, as shown in Figure 5, further demonstrating its robustness and generalization.

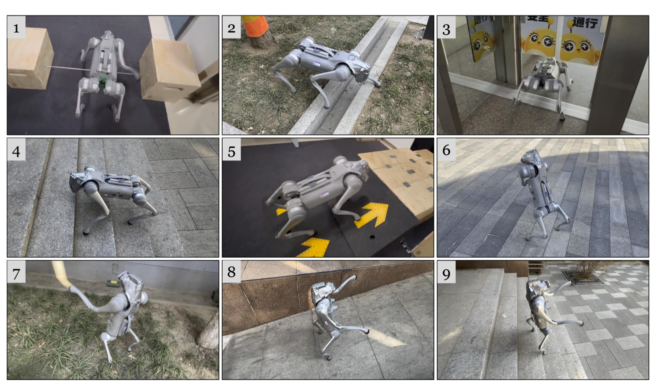

Fig. 5: **Real-world experiments over multiple terrains and gaits:** 1. Bar (Quad), 2. Pit (Quad), 3. Baffle (Quad), 4. Stair (Quad), 5. Slope (Quad), 6. Stand up (Bip), 7. Walk (Bip), 8. Slope (Bip), 9. Stair (Bip).

C. Gradient Conflict Alleviation

In order to unveil whether gradient conflict can be reduced by applying mixture of experts, we conducted gradient conflict experiments. We resume from the checkpoints after pretraining for 15000 epochs and run the training process of multitask for 500 epochs of 4096 quadrupedal robots. We average the gradient throughout the process. We consider two metrics to measure the gradient conflict of robot locomotion. **1) Cosine Similarity:** We compute the normalized dot product of all parameters' gradients of different tasks. Smaller cosine similarity indicates larger gradient conflicts. **2) Negative gradient ratio:** We compute negative gradient update ratio for each pair of task gradients. Larger negative gradient ratio indicates larger gradient conflict. We test the gradient conflict between 5 different tasks: quadrupedal bar crossing, quadrupedal baffle crawling, quadrupedal stair walking, bipedal plane walking and bipedal slope walking.

TABLE III: **Cosine similarity of gradients of different tasks.** Left represents MoE policy and right represents standard policy.

MoE/Standard	Gradient Cosine Similarity ↑				
	Bar (q)	Baffle (q)	Stair (q)	Slope Up (b)	Slope down (b)
Bar (q)	-	**0.519**/0.474	**0.606**/0.592	**0.278**/-0.132	**0.091**/-0.128
Baffle (q)	-	-	0.369/**0.384**	**0.062**/-0.091	**0.061**/-0.101
Stair (q)	-	-	-	**0.046**/-0.023	**0.052**/0.015
Slope up (b)	-	-	-	-	**0.806**/0.709
Slope down (b)	-	-	-	-	-

TABLE IV: **Negative entry ratio of MoE and Standard policy on different tasks.** Left represents MoE policy and right represents standard policy.

MoE/Standard	Gradient Negative Entries (%) ↓				
	Bar (q)	Baffle (q)	Stair (q)	Slope Up (b)	Slope down (b)
Bar (q)	-	**35.72**/37.33	32.67/**32.62**	**45.50**/ 55.12	**49.83**/50.80
Baffle (q)	-	-	39.90/**38.52**	**49.86**/55.91	**49.91**/51.68
Stair (q)	-	-	-	**49.52**/50.15	**50.04**/50.34
Slope up (b)	-	-	-	-	**23.17**/30.91
Slope down (b)	-	-	-	-	-

As shown in Table III and Table IV, the MoE policy significantly reduces gradient conflict between bipedal and quadrupedal tasks. It also minimizes gradient conflict even between quadrupedal tasks that require fundamentally different skills, such as quadrupedal bar crossing and quadrupedal baffle crawling.

D. Training Performance

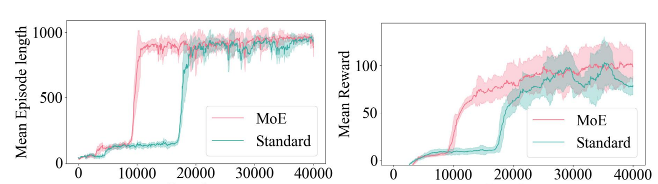

Fig. 6: **Training curve of our multitask policy in the pretraining stage.**

In terms of training performance, we focus on the mean reward and mean episode length. The mean reward reflects the policy's ability to exploit the environment, while the mean episode length indicates how well the robot learns to stand and walk. We present the training curve during the plane pretraining stage, where the robot learns both bipedal and quadrupedal plane walking. As shown in Figure 6, our MoE policy outperforms the standard policy with similar total parameters across both metrics.

斜坡行走的相对简单性。此外，没有MoE的策略难以有效穿越具有挑战性的多任务地形设置，因为它受到梯度冲突的显著影响，如第IV-C小节所讨论的。关于RMA，我们遵循其原始实现，该实现仅使用MLP主干网络和CNN编码器。这种设计选择导致其在多个具有挑战性的地形上表现欠佳。

2) 真实世界实验：我们将MoE策略零样本部署到真实机器人上，并开展真实世界实验。我们测试了包含所有挑战性任务的混合地形，以及每种单独的地形。对于混合地形，我们的机器人首先需要以四足步态依次跨越20厘米栏杆、22厘米挡板、15厘米楼梯、20厘米坑和30度斜坡。随后，它接收到双足指令，站立起来，走上30度斜坡，转身并走下坡。对于每项测试，我们进行20次试验并记录平均成功率。我们还针对单一任务测试了不同的策略。

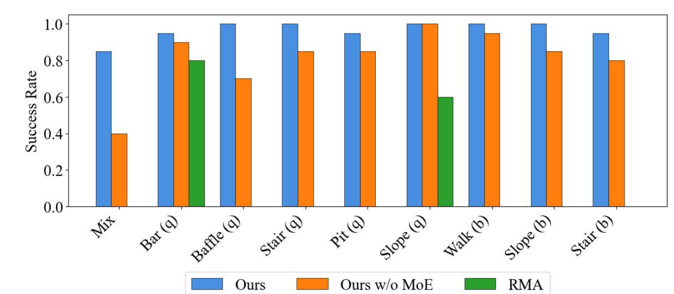

图4：**真实世界中多种地形和步态下的成功率。**

如图4所示，我们的方法在所有类型的任务上都取得了更好的性能，并在混合地形中表现出显著更高的成功率。此外，我们还在更多户外环境中进行了实验，如图5所示，进一步证明了其鲁棒性和泛化性。

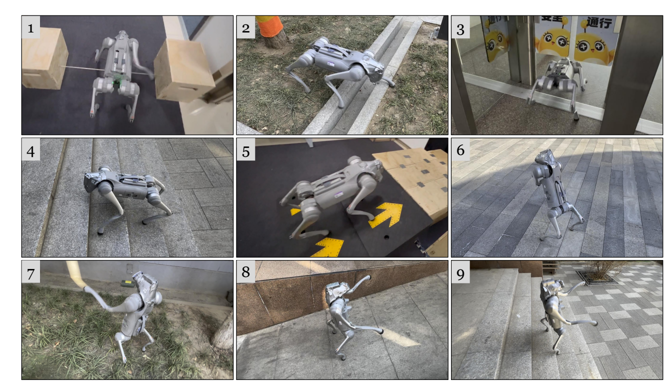

图5：**多种地形和步态下的真实世界实验：**1. 栏杆（四足），2. 坑（四足），3. 挡板（四足），4. 楼梯（四足），5. 坡道（四足），6. 站立（双足），7. 行走（双足），8. 坡道（双足），9. 楼梯（双足）。

C. 梯度冲突缓解

为了揭示应用专家混合是否能减少梯度冲突，我们进行了梯度冲突实验。我们从预训练15000轮后的检查点恢复，并运行4096个四足机器人的多任务训练过程500轮。我们平均整个过程中的梯度。我们考虑两个指标来衡量机器人运动的梯度冲突。**1) 余弦相似度：**我们计算不同任务所有参数梯度的归一化点积。较小的余弦相似度表示较大的梯度冲突。**2) 负梯度比率：**我们计算每对任务梯度的负梯度更新比率。较大的负梯度比率表示较大的梯度冲突。我们测试了5个不同任务之间的梯度冲突：四足栏杆跨越、四足挡板爬行、四足楼梯行走、双足平地行走和双足斜坡行走。

表III：**不同任务梯度的余弦相似度**。左侧表示MoE策略，右侧表示标准策略。

MoE/标准	梯度余弦相似度 ↑				
	栏杆 (q)	挡板 (q)	Stair (q)	Slope Up (b)	Slope down (b)
Bar (q)	-	**0.519**/0.474	**0.606**/0.592	**0.278**/-0.132	**0.091**/-0.128
Baffle (q)	-	-	0.369/**0.384**	**0.062**/-0.091	**0.061**/-0.101
Stair (q)	-	-	-	**0.046**/-0.023	**0.052**/0.015
上坡 (b)	-	-	-	-	**0.806**/0.709
下坡 (b)	-	-	-	-	-

表IV：**MoE与标准策略在不同任务上的负项比例**。左侧代表MoE策略，右侧代表标准策略。

MoE/标准	梯度负项（%） ↓				
	栏杆 (q)	挡板 (q)	Stair (q)	Slope Up (b)	Slope down (b)
Bar (q)	-	**35.72**/37.33	32.67/**32.62**	**45.50**/ 55.12	**49.83**/50.80
Baffle (q)	-	-	39.90/**38.52**	**49.86**/55.91	**49.91**/51.68
楼梯 (q)	-	-	-	**49.52**/50.15	**50.04**/50.34
上坡 (b)	-	-	-	-	**23.17**/30.91
下坡 (b)	-	-	-	-	-

如表III 和表IV所示，MoE策略显著降低了双足行走与四足行走任务之间的梯度冲突。它甚至能最小化那些需要根本不同技能的四足行走任务之间的梯度冲突，例如四足横杆穿越和四足挡板爬行。

D. 训练性能

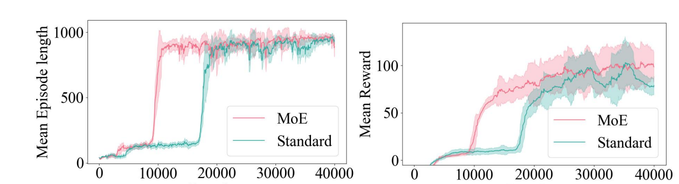

图6：**我们的多任务策略在预训练阶段的训练曲线。**

在训练性能方面，我们关注平均奖励和平均回合长度。平均奖励反映了策略利用环境的能力，而平均回合长度则表明机器人学习站立和行走的效果。我们展示了平面预训练阶段的训练曲线，在该阶段机器人学习双足行走和四足平面行走。如图6所示，我们的MoE策略在总参数相近的情况下，在这两项指标上均优于标准策略。

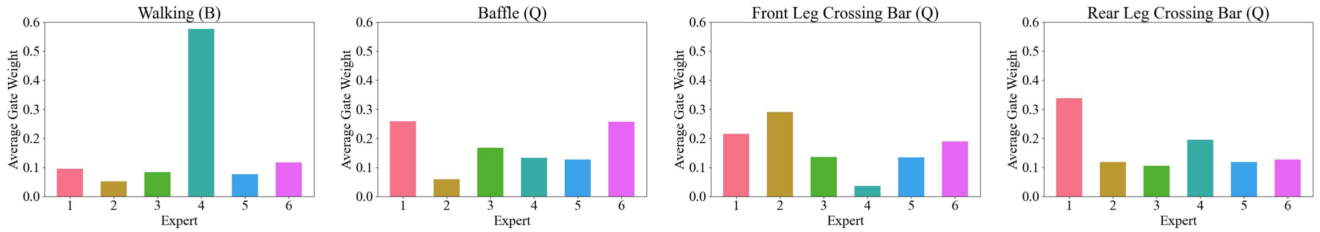

Fig. 7: **Expert usage in different tasks.** From left to right is: Bipedal walking, Quadrupedal Baffle crawling, Front Leg Crossing Bars and Rear Leg Crossing Bars.

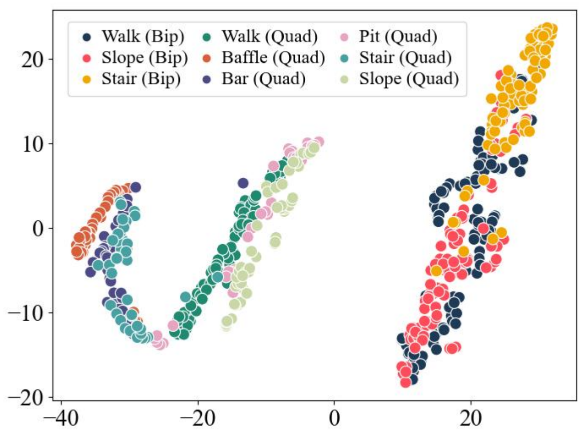

Fig. 8: **t-SNE result of gating network output on different terrains and gaits.**

E. Expert Specialization Analysis

We conduct both qualitative and quantitative experiments to analyze the emergent composition of skills. As shown in Figure 7, we plot the mean weight of different experts across various tasks. It is evident that the distribution of gating weights varies significantly from task to task, demonstrating the expertise and differentiation of various experts.

To further analyze the composition of different experts across various tasks, we use t-SNE to visualize the output of the gating network for different tasks (i.e., the weight of each expert). As shown in Figure 8, the bipedal and quadrupedal tasks form distinct clusters. Quadrupedal slope walking and quadrupedal pit crossing tasks are performed using a gait similar to quadrupedal plane walking, and thus, they cluster closely together. In contrast, bar crossing, baffle crawling, and stair climbing exhibit a gait that differs more significantly, resulting in them clustering further apart.

F. Skill Composition

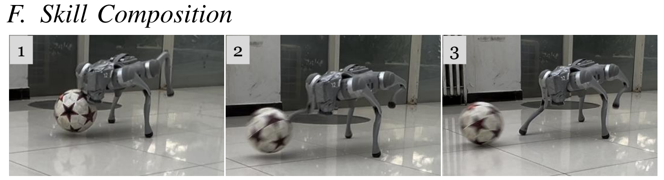

Fig. 9: **Manually designed new dribbling gait by selecting two experts.**

During our training, we found that our experts not only specialize and cooperate across different tasks, but they also emerge with specific, human-interpretable skills. We discovered that one expert specializes in balancing, which helps lift the robot's body but limits its agility. Another expert is responsible for lifting one of the front legs to perform crossing tasks, enabling the robot to execute basic movement skills. By selecting the balancing expert and the crossing expert, we are able to zero-shot transfer to a new dribbling pattern. In this gait, we select the two experts mentioned above, manually double the gating weight of the crossing expert, and mask out all other experts. As shown in Figure 9, the new dribbling gait allows the robot to walk effectively while periodically using one of its front legs to kick the ball. This skill composition results from the automatic skill decomposition and interpretability inherent in the MoE architecture. In contrast, a standard neural network would function as a black box, lacking such interpretable skill decomposition.

G. Additional Experiment

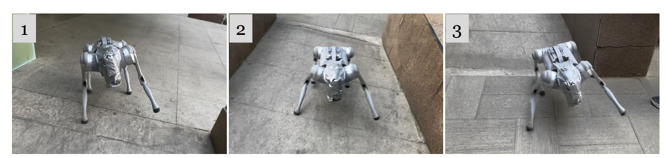

Fig. 10: **`MoE-Loco` can quickly adapt to a three-footed gait by training a new expert.** 1) ground plane, 2) slope up, and 3) slope down.

We conduct an adaptation learning experiment to demonstrate how our pretrained experts can be recomposed and adapted to new tasks. In this experiment, we design the robot to walk on three feet. We introduce a newly initialized expert, freeze the parameters of the original experts, and update only the gating network. As shown in Figure 10, the robot can walk on both flat ground and a slope using only three feet. The newly added expert only needs to learn how to lift one leg, while leveraging the walking and slope capabilities of the original experts.

V. CONCLUSION

We introduced MoE-Loco, a multitask locomotion framework that utilizes a Mixture of Experts architecture to train a single policy for quadrupedal robots. Our approach effectively mitigates gradient conflicts, leading to improved training efficiency and task performance. Through extensive evaluations in both simulated and real-world environments, we demonstrated the capability of our method to handle a variety of terrains and gaits. Future work will explore extending this approach to incorporate sensory perception such as camera and Lidar to enhance adaptability in more complex tasks.

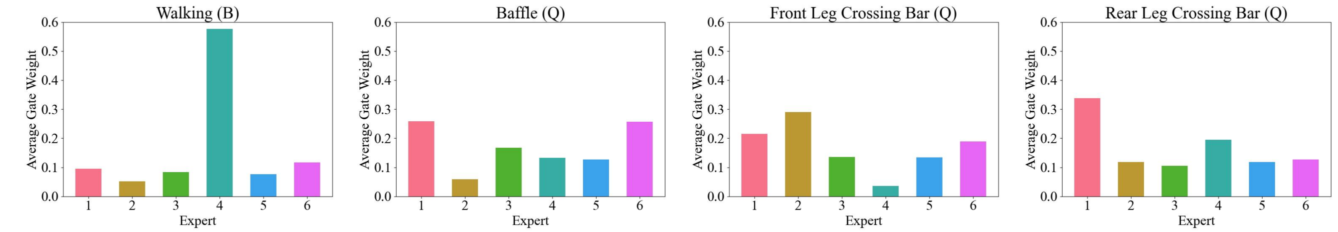

图7：**不同任务中的专家使用情况**。从左到右依次为：双足行走、四足挡板爬行、前腿跨栏和后腿跨栏。

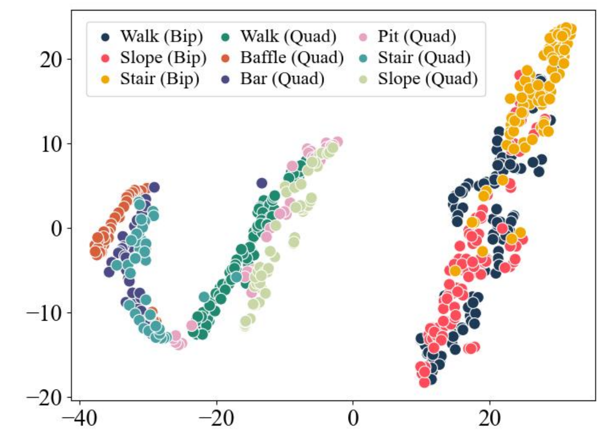

图8：**门控网络输出在不同地形和步态上的t-SNE结果。**

E. 专家特化分析

我们进行了定性和定量实验来分析技能的涌现组合。如图7所示，我们绘制了不同专家在各种任务中的平均权重。显然，门控权重的分布因任务而异，这证明了各专家的专业性和差异化。

为了进一步分析不同任务中各专家的组合情况，我们使用t-SNE对不同任务的门控网络输出（即每个专家的权重）进行可视化。如图8所示，双足行走和四足行走任务形成了不同的聚类。四足斜坡行走和四足跨坑任务采用与四足平面行走相似的步态，因此它们聚类紧密。相比之下，跨栏、挡板爬行和爬楼梯所展现的步态差异较大，导致它们聚类距离较远。

F. 技能组合

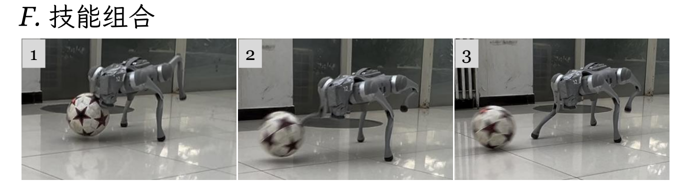

图9：**通过选择两个专家手动设计的新运球步态。**

在训练过程中，我们发现我们的专家不仅在不同任务间实现了专业化和协作，还涌现出特定且人类可解释的技能。我们发现，一位专家专门负责平衡，这有助于提升机器人的身体，但限制了其敏捷性。另一位专家则负责抬起其中一条前腿以执行跨越任务，使机器人能够执行基本运动技能。通过选择平衡专家和跨越专家，我们能够零样本迁移到一种新的运球模式。在这种步态中，我们选择上述两位专家，手动将跨越专家的门控权重加倍，并屏蔽所有其他专家。如图9所示，这种新的运球步态使机器人能够有效行走，同时定期使用其一条前腿踢球。这种技能组合源于MoE架构中固有的自动技能分解和可解释性。相比之下，标准神经网络将充当黑盒，缺乏这种可解释的技能分解。

G. 附加实验

图10：**`MoE-Loco` 可以通过训练新专家快速适应三足步态。** 1) 平坦地面，2) 上坡，3) 下坡。

我们进行了一项适应学习实验，以展示预训练专家如何被重新组合并适应新任务。在此实验中，我们设计机器人以三足行走。我们引入一个新初始化的专家，冻结原始专家的参数，仅更新门控网络。如图10所示，机器人仅用三足便可在平坦地面和坡道上行走。新添加的专家只需学习如何抬起一条腿，同时利用原始专家的行走和坡道适应能力。

V. 结论

我们引入了MoE-Loco，这是一个利用专家混合架构训练四足机器人单一策略的多任务运动框架。我们的方法有效缓解了梯度冲突，从而提升了训练效率和任务性能。通过在仿真和真实环境中的广泛评估，我们证明了该方法能够应对多种地形和步态。未来的工作将探索扩展该方法以融合相机和激光雷达等感知，从而增强在更复杂任务中的适应性。

APPENDIX

A. Reward Functions

The robots in bipedal gait receives different reward to the robots in quadrupedal gait. Detailed explanation is shown in Table V.

TABLE V: Reward functions

Type	Item	Formula	Weight
Quadrupedal Tracking	Tracking lin vel	$\exp\left(-\frac{\lVert \mathbf{v}_{c,xy}-\mathbf{v}_{b,xy}\rVert^2}{\sigma}\right)$	7.0
	Tracking ang vel	$\exp\left(-\frac{(\mathbf{c}_{yaw}-\omega_{yaw})^2}{\sigma}\right)$	2.5
	Termination	-1	-1.0
	Alive	1	1.0
Quadrupedal Regularization	Joint pos	$(\mathbf{q}-\mathbf{q}_{default})^2$	-0.05
	Joint vel	$\lVert\dot{\mathbf{q}}\rVert_2$	-0.002
	Joint acc	$\lVert\ddot{\mathbf{q}}\rVert_2$	-2×10^{-6}
	Ang vel stability	$\left(\lVert\omega_{t,x}\rVert_2+\lVert\omega_{t,y}\rVert_2\right)$	-0.2
	Feet in air	$\prod_{i=1}^{N}\mathbb{I}\left\{\mathbf{F}^{foot}_{i,z}<1\right\}+\sum_{i=1}^{N}\mathbb{I}\left\{\mathbf{F}^{foot}_{i,z}<1\right\}\mathbb{I}\left\{\mathbf{F}^{calf}_{i,z}\geq1\right\}$	-0.05
	Front hip pos	$\sum_{i\in\mathcal{I}_{front\ hip}}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.2
	Rear hip pos	$\sum_{i\in\mathcal{I}_{rear\ hip}}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.5
	Base height	$\left(\mathbf{z}_{base}-\frac{1}{\mathbf{N}}\sum_{i=1}^{\mathbf{N}}\mathbf{h}_i-\mathbf{h}_{target}\right)^2$	-0.1
	Balance	$\lVert\mathbf{F}_{feet,0}+\mathbf{F}_{feet,2}-\mathbf{F}_{feet,1}-\mathbf{F}_{feet,3}\rVert_2$	-2×10^{-5}
	Joint limit	$\left\lVert\left(\mathbf{q}<\mathbf{q}_{min}\right)\vee\left(\mathbf{q}>\mathbf{q}_{max}\right)\right\rVert_1$	-0.01
	Torque exceed limit	$\sum_{i=1}^{\mathbf{N}_{sub}}\sum_{j=1}^{12}\max\left\{\lvert\tau_{ij}\rvert-\tau_j^{limit},0\right\}$	-2.0
Bipedal Stand	Orientation	$\left(0.5\cos\theta+0.5\right)^2,\quad\theta=\arccos\left(\frac{\mathbf{g}\cdot\mathbf{t}}{\lVert\mathbf{g}\rVert\lVert\mathbf{t}\rVert}\right)$	1.0
	Base height linear	$\min\left(\max\left(\frac{\mathbf{z}_{root}-T_{min}}{T_{max}-T_{min}},0\right),1\right)$	0.8
Bipedal Tracking	Tracking lin vel	$\exp\left(-\frac{\lVert\mathbf{v}_c-\mathbf{v}_a\rVert^2}{\sigma}\right)\mathbb{I}\left\{\cos\theta>0.95\right\}\frac{\mathbf{z}_{base}-\mathbf{s}_{low}}{\mathbf{s}_{high}-\mathbf{s}_{low}}$	3.0
	Tracking ang vel	$\exp\left(-\frac{(\mathbf{c}_{roll}-\omega_{roll})^2}{\sigma_{ang}}\right)\mathbb{I}\left\{\cos\theta>0.95\right\}\frac{\mathbf{z}_{base}-\mathbf{s}_{low}}{\mathbf{s}_{high}-\mathbf{s}_{low}}$	2.5
	Termination	-1	-1.0
	Alive	1	1.0
Bipedal Regularization	Rear air	$\prod_{i\in\mathbf{R}}\mathbb{I}\left\{\mathbf{f}^{rear}_{i,z}<1\right\}+\sum_{i\in\mathbf{R}}\mathbb{I}\left\{\mathbf{f}^{rear}_{i,z}<1\right\}\mathbb{I}\left\{\mathbf{f}^{rear-calf}_{i,z}\geq1\right\}$	-0.5
	Front hip pos	$\sum_{i\in\mathbf{F}_H}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.1
	Rear hip pos	$\sum_{i\in\mathbf{R}_H}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.18
	Rear pos balance	$\left\lVert\mathbf{q}_i^{rear\,left}-\mathbf{q}_i^{rear\,right}\right\rVert_2$	-0.05
	Front joint pos	$\mathbf{1}\left\{\mathbf{t}>\mathbf{T}_{allow}\right\}\sum_{i\in\mathbf{F}}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.2
	Front joint vel	$\mathbf{1}\left\{\mathbf{t}>\mathbf{T}_{allow}\right\}\sum_{i\in\mathbf{F}}\dot{\mathbf{q}}_i^2$	-1×10^{-3}
	Front joint acc	$\mathbf{1}\left\{\mathbf{t}>\mathbf{T}_{allow}\right\}\sum_{i\in\mathbf{F}}\left(\frac{\Delta\dot{\mathbf{q}}_i}{\Delta\mathbf{t}}\right)^2$	-2×10^{-6}
	Legs energy substeps	$\frac{1}{\mathbf{N}_s}\sum_{j=1}^{\mathbf{N}_s}\sum_{i=1}^{\mathbf{N}_{dof}}\left(\tau_{ij}\,\dot{\mathbf{q}}_{ij}\right)^2$	-1×10^{-6}
	Torque exceed limits	$\sum_j\sum_i\max\left\{\lvert\tau_{ij}\rvert-\tau_i^{limit},0\right\}$	-2.0
	Joint limits	$\frac{1}{\mathbf{N}_s}\sum_j\sum_i\mathbb{I}\left\{\mathbf{q}_{ij}\notin[\mathbf{q}_i^{min},\mathbf{q}_i^{max}]\right\}$	-0.06
	Collision	$\sum_k\mathbb{I}\left\{\lVert\mathbf{f}_k\rVert>1\right\}$	-2.0
	Action rate	$\lVert\mathbf{a}_i^{last}-\mathbf{a}_i\rVert_2$	-0.03
	Joint vel	$\lVert\dot{\mathbf{q}}\rVert_2$	-2×10^{-3}
	Joint acc	$\lVert\ddot{\mathbf{q}}\rVert_2$	-3×10^{-6}

B. Algorithm Pseudocode

Algorithm 1 Training Stage 1

for total iteration **do**
  Initialize rollout buffer $\mathcal{D}\leftarrow\emptyset$
  for num steps **do**
    $\mathbf{z}_t\leftarrow\mathrm{Enc}(\mathbf{i}_t)$
    $\hat{\mathbf{l}}_t\leftarrow[\mathrm{Estimator}(\mathbf{p}_t,\mathbf{c}_t),\mathbf{p}_t]$
    $\mathbf{l}_t\leftarrow[\mathbf{z}_t,\mathbf{e}_t,\mathbf{p}_t]$
    $\mathbf{h}_t\leftarrow\mathrm{LSTM}([\mathbf{l}_t,\mathbf{c}_t])$
    $\hat{\mathbf{g}}\leftarrow\mathrm{softmax}(g(\mathbf{h}_t))$
    $\mathbf{a}_t\leftarrow\sum_{i=1}^{N}\hat{\mathbf{g}}_i\cdot f_i(\mathbf{h}_t)$
    Execute $\mathbf{a}_t$, observe reward r_t and next state
    Reset if terminated
    $t\leftarrow t+1$
    Store rollout in $\mathcal{D}$
  end for
  Compute PPO losses: L_{surro}, L_{value}
  Compute reconstruction loss:
    $L_{recon}=\sum_{\hat{\mathbf{l}}_i,\mathbf{l}_i\in\mathcal{D}}\left\lVert\hat{\mathbf{l}}_i-\mathbf{l}_i\right\rVert^2$
  $L=L_{surro}+L_{value}+L_{recon}$
  Update policy and value network
end for
return Oracle Policy

Algorithm 2 Training Stage 2

Copy parameters from Oracle Policy
for total iteration **do**
  Initialize rollout buffer $\mathcal{D}\leftarrow\emptyset$
  for num steps **do**
    $\hat{\mathbf{l}}_t\leftarrow[\mathrm{Estimator}(\mathbf{p}_t,\mathbf{c}_t),\mathbf{p}_t]$
    $\mathbf{l}_t\leftarrow[\mathbf{z}_t,\mathbf{e}_t,\mathbf{p}_t]$
    $\mathbf{P}_t\leftarrow\alpha^t$
    $\bar{\mathbf{l}}_t\leftarrow\mathrm{Probability\ Selection}(\mathbf{P}_t,\hat{\mathbf{l}}_t,\mathbf{l}_t)$
    $\mathbf{h}_t\leftarrow\mathrm{LSTM}([\bar{\mathbf{l}}_t,\mathbf{c}_t])$
    $\hat{\mathbf{g}}\leftarrow\mathrm{softmax}(g(\mathbf{h}_t))$
    $\mathbf{a}_t\leftarrow\sum_{i=1}^{N}\hat{\mathbf{g}}_i\cdot f_i(\mathbf{h}_t)$
    Execute $\mathbf{a}_t$, observe reward r_t and next state
    Reset if terminated
    $t\leftarrow t+1$
    Store rollout in $\mathcal{D}$
  end for
  Compute PPO losses: L_{surro}, L_{value}
  Compute reconstruction loss:
    $L_{recon}=\sum_{\hat{\mathbf{l}}_i,\mathbf{l}_i\in\mathcal{D}}\left\lVert\hat{\mathbf{l}}_i-\mathbf{l}_i\right\rVert^2$
  $L=L_{surro}+L_{value}+L_{recon}$
  Update policy and value network
end for
return Final Policy

C. Network Architecture Details

The architecture of different modules used in our experiment is shown in Table VI.

TABLE VI: Network architecture details

Network	Type	Dims
Actor RNN	LSTM	[256]
Critic RNN	LSTM	[256]
Estimator Module	LSTM	[256]
Estimator Latent Encoder	MLP	[256, 128]
Implicit Encoder	MLP	[32, 16]
Expert Head	MLP	[256, 128, 128]
Standard Head	MLP	[640, 640, 128]
Gating Network	MLP	[128]

D. Domain Randomization

We introduce dynamic randomization to ensure our policy can be safely transfered to real environment. The random ranges are shown in the Table VII.

TABLE VII: Domain randomization

Parameters	Range	Unit
Base mass	[1, 3]	kg
Mass position of X axis	[-0.2, 0.2]	m
Mass position of Y axis	[-0.1, 0.1]	m
Mass position of Z axis	[-0.05, 0.05]	m
Friction	[0, 2]	-
Initial joint positions	[0.5, 1.5] × nominal value	rad
Motor strength	[0.9, 1.1] × nominal value	-
Proprioception latency	[0.005, 0.045]	s

Gaussian noise are added to the input observation to simulate the noise in the real world, as shown in Table VIII.

TABLE VIII: Gaussian noise

Observation	Gaussian Noise Amplitude	Unit
Linear velocity	0.05	m/s
Angular velocity	0.2	rad/s
Gravity	0.05	m/s^2
Joint position	0.01	rad
Joint velocity	1.5	rad/s

附录

A. 奖励函数

双足步态中的机器人所获得的奖励与四足步态中的机器人不同。详细说明见表V。

表V：奖励函数

Type	Item	公式	权重
四足跟踪	跟踪线速度	$\exp\left(-\frac{\lVert \mathbf{v}_{c,xy}-\mathbf{v}_{b\,xy}\rVert^2}{\sigma}\right)$	7.0
	跟踪角速度	$\exp\left(-\frac{(\mathbf{c}_{yaw}-\omega_{yaw})^2}{\sigma}\right)$	2.5
	终止	-1	-1.0
	存活	1	1.0
四足正则化	关节位置	$(\mathbf{q}-\mathbf{q}_{default})^2$	-0.05
	关节速度	$\lVert\dot{\mathbf{q}}\rVert_2$	-0.002
	关节加速度	$\lVert\ddot{\mathbf{q}}\rVert_2$	-2×10^{-6}
	角速度稳定性	$\left(\lVert\omega_{t,x}\rVert_2+\lVert\omega_{t,y}\rVert_2\right)$	-0.2
	脚离地	$\prod_{i=1}^{N}\mathbb{I}\left\{\mathbf{F}^{foot}_{i,z}<1\right\}+\sum_{i=1}^{N}\mathbb{I}\left\{\mathbf{F}^{foot}_{i,z}<1\right\}\mathbb{I}\left\{\mathbf{F}^{calf}_{i,z}\geq1\right\}$	-0.05
	前髋位置	$\sum_{i\in我_{front\ hip}}\left(问_i-问_i^{default}\right)^2$	-0.2
	后髋位置	$\sum_{i\in我_{rear\ hip}}\left(问_i-问_i^{default}\right)^2$	-0.5
	基座高度	$\left(\mathbf{z}_{base}-\frac{1}{\mathbf{N}}\sum_{i=1}^{\mathbf{N}}\mathbf{h}_i-\mathbf{h}_{target}\right)^2$	-0.1
	平衡	$\lVert\mathbf{F}_{feet,0}+\mathbf{F}_{feet,2}-\mathbf{F}_{feet,1}-\mathbf{F}_{feet,3}\rVert_2$	-2×10^{-5}
	关节限位	$\left\lVert\left(\mathbf{q}<\mathbf{q}_{min}\right)\vee\left(\mathbf{q}>\mathbf{q}_{max}\right)\right\rVert_1$	-0.01
	扭矩超限	$\sum_{i=1}^{\mathbf{N}_{sub}}\sum_{j=1}^{12}最大\left\{\lvert\tau_{ij}\rvert-\tau_j^{limit},0\right\}$	-2.0
双足站立	朝向	$\left(0。5\cos\theta+0。5\right)^2,\quad\theta=\arccos\left(\frac{\mathbf{g}\cdot\mathbf{t}}{\lVert\mathbf{g}\rVert\lVert\mathbf{t}\rVert}\right)$	1.0
	基座高度线性	$最小\left(最大\left(\frac{\mathbf{z}_{root}-T_{min}}{T_{max}-T_{min}},0\right),1\right)$	0.8
双足跟踪	跟踪线速度	$指数\left(-\frac{\lVert\mathbf{v}_c-\mathbf{v}_a\rVert^2}{\sigma}\right)\mathbb{I}\left\{余弦\,\theta>0.95\right\}\frac{\mathbf{z}_{base}-\mathbf{s}_{low}}{\mathbf{s}_{high}-\mathbf{s}_{low}}$	3.0
	跟踪角速度	$指数\left(-\frac{(\mathbf{c}_{roll}-\omega_{roll})^2}{\sigma_{ang}}\right)\mathbb{I}\left\{余弦\,\theta>0.95\right\}\frac{\mathbf{z}_{base}-\mathbf{s}_{low}}{\mathbf{s}_{high}-\mathbf{s}_{low}}$	2.5
	终止	-1	-1.0
	存活	1	1.0
双足正则化	后腿腾空	$\prod_{i\in\mathbf{R}}\mathbb{I}\left\{\mathbf{f}^{rear}_{i,z}<1\right\}+\sum_{i\in\mathbf{R}}\mathbb{I}\left\{\mathbf{f}^{rear}_{i,z}<1\right\}\mathbb{I}\left\{\mathbf{f}^{rear-calf}_{i,z}\geq1\right\}$	-0.5
	前髋位置	$\sum_{i\in\mathbf{F}_H}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.1
	后髋位置	$\sum_{i\in\mathbf{R}_H}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.18
	后位置平衡	$\left\lVert\mathbf{q}_i^{rear\,left}-\mathbf{q}_i^{rear\,right}\right\rVert_2$	-0.05
	前关节位置	$\mathbf{1}\left\{\mathbf{t}>\mathbf{T}_{allow}\right\}\sum_{i\in\mathbf{F}}\left(\mathbf{q}_i-\mathbf{q}_i^{default}\right)^2$	-0.2
	前关节速度	$\mathbf{1}\left\{\mathbf{t}>\mathbf{T}_{allow}\right\}\sum_{i\in\mathbf{F}}\dot{\mathbf{q}}_i^2$	-1×10^{-3}
	前关节加速度	$\mathbf{1}\left\{\mathbf{t}>\mathbf{T}_{allow}\right\}\sum_{i\in\mathbf{F}}\left(\frac{\Delta\dot{\mathbf{q}}_i}{\Delta\mathbf{t}}\right)^2$	-2×10^{-6}
	腿部能量子步	$\frac{1}{\mathbf{N}_s}\sum_{j=1}^{\mathbf{N}_s}\sum_{i=1}^{\mathbf{N}_{dof}}\left(\tau_{ij}\,\dot{\mathbf{q}}_{ij}\right)^2$	-1×10^{-6}
	扭矩超限	$\sum_j\sum_i最大\left\{\lvert\tau_{ij}\rvert-\tau_i^{limit},0\right\}$	-2.0
	关节限位	$\frac{1}{\mathbf{N}_s}\sum_j\sum_i\mathbb{I}\left\{\mathbf{q}_{ij}\notin[\mathbf{q}_i^{min},\mathbf{q}_i^{max}]\right\}$	-0.06
	碰撞	$\sum_k\mathbb{I}\left\{\lVert\mathbf{f}_k\rVert>1\right\}$	-2.0
	动作速率	$\lVert\mathbf{a}_i^{last}-\mathbf{a}_i\rVert_2$	-0.03
	关节速度	$\lVert\dot{\mathbf{q}}\rVert_2$	-2×10^{-3}
	关节加速度	$\lVert\ddot{\mathbf{q}}\rVert_2$	-3×10^{-6}

B. 算法伪代码

算法1 训练阶段1

对于 总迭代次数**执行**
  初始化回放缓冲区 D $\leftarrow\emptyset$
  对于 步数**执行**
    $\mathbf{z}_t\leftarrow\mathrm{Enc}(\mathbf{i}_t)$
    $\hat{\mathbf{l}}_t\leftarrow$ [估计器($\mathbf{p}_t,\mathbf{c}_t$), $\mathbf{p}_t$]
    $\mathbf{l}_t\leftarrow[\mathbf{z}_t,\mathbf{e}_t,\mathbf{p}_t]$
    $\mathbf{h}_t\leftarrow\mathrm{LSTM}([\mathbf{l}_t,\mathbf{c}_t])$
    $\hat{\mathbf{g}}\leftarrow\mathrm{softmax}(g(\mathbf{h}_t))$
    $\mathbf{a}_t\leftarrow\sum_{i=1}^{N}\hat{\mathbf{g}}_i\cdot f_i(\mathbf{h}_t)$
    执行 $\mathbf{a}_t$，观察奖励 r_t 和下一状态
    如果终止则重置
    $t\leftarrow t+1$
    将轨迹存储在D中
  结束循环
  计算PPO损失：L_{surro}, L_{value}
  计算重建损失：
    $L_{recon}=\sum_{\hat{\mathbf{l}}_i,\mathbf{l}_i\in\mathcal{D}}\left\lVert\hat{\mathbf{l}}_i-\mathbf{l}_i\right\rVert^2$
  $L=L_{surro}+L_{value}+L_{recon}$
  更新策略和价值网络
结束循环
返回Oracle策略

算法2 训练阶段2

从Oracle策略复制参数
对于 总迭代次数**执行**
  初始化回放缓冲区 D $\leftarrow\emptyset$
  对于 步数**执行**
    $\hat{\mathbf{l}}_t\leftarrow$ [估计器($\mathbf{p}_t,\mathbf{c}_t$), $\mathbf{p}_t$]
    $\mathbf{l}_t\leftarrow[\mathbf{z}_t,\mathbf{e}_t,\mathbf{p}_t]$
    $\mathbf{P}_t\leftarrow\alpha^t$
    $\bar{\mathbf{l}}_t\leftarrow$ 概率选择($\mathbf{P}_t,\hat{\mathbf{l}}_t,\mathbf{l}_t$)
    $\mathbf{h}_t\leftarrow\mathrm{LSTM}([\bar{\mathbf{l}}_t,\mathbf{c}_t])$
    $\hat{\mathbf{g}}\leftarrow\mathrm{softmax}(g(\mathbf{h}_t))$
    $\mathbf{a}_t\leftarrow\sum_{i=1}^{N}\hat{\mathbf{g}}_i\cdot f_i(\mathbf{h}_t)$
    执行 $\mathbf{a}_t$，观察奖励 r_t 和下一状态
    若终止则重置
    $t\leftarrow t+1$
    将轨迹存储到D中
  结束循环
  计算PPO损失：L_{surro}, L_{value}
  计算重建损失：
    $L_{recon}=\sum_{\hat{\mathbf{l}}_i,\mathbf{l}_i\in\mathcal{D}}\left\lVert\hat{\mathbf{l}}_i-\mathbf{l}_i\right\rVert^2$
  $L=L_{surro}+L_{value}+L_{recon}$
  更新策略和价值网络
结束循环
返回最终策略

C. 网络架构细节

我们实验中使用的不同模块的架构如表VI所示。表VI：网络架构细节

网络	Type	Dims
Actor RNN	LSTM	[256]
Critic RNN	LSTM	[256]
估计器模块	LSTM	[256]
估计器潜在编码器	MLP	[256, 128]
隐式编码器	MLP	[32, 16]
专家头	MLP	[256, 128, 128]
标准头	MLP	[640, 640, 128]
门控网络	MLP	[128]

D. 域随机化

我们引入动态随机化，以确保我们的策略能够安全地迁移到真实环境中。随机范围如表VII所示。

表VII：域随机化

参数	范围	Unit
基座质量	[1, 3]	kg
X轴质量位置	[-0.2, 0.2]	m
Y轴质量位置	[-0.1, 0.1]	m
Z轴质量位置	[-0.05, 0.05]	m
摩擦力	[0, 2]	-
初始关节位置	[0.5, 1.5] × 标称值	rad
电机强度	[0.9, 1.1] × 标称值	-
本体感觉延迟	[0.005, 0.045]	s

在输入观测中添加高斯噪声，以模拟真实世界中的噪声，如表VIII所示表VIII。

表VIII：高斯噪声

观测	高斯噪声幅度	Unit
线速度	0.05	m/s
角速度	0.2	rad/s
重力	0.05	m/s^2
关节位置	0.01	rad
关节速度	1.5	rad/s

E. Training Details

In order to prevent legs dragging on the floor, and achieve a robust performance on uneven terrains, we apply fractal noise [7] to the ground. We use maximum noise scale $z_{max} = 0.1$. We use PPO as our reinforcement learning algorithm. The hyperparameters are shown in Table IX.

TABLE IX: PPO hyperparameters

Hyperparameter	Value
clip min std	0.05
clip param	0.2
gamma	0.99
lam	0.95
desired kl	0.01
entropy coef	0.01
learning rate	0.001
max grad norm	1
num mini batch	4
num steps per env	24

References

[1] K. N. Kumar, I. Essa, and S. Ha, "Cascaded compositional residual learning for complex interactive behaviors," *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 4601–4608, 2023. 1, 2

[2] A. Klipfel, N. Sontakke, R. Liu, and S. Ha, "Learning a single policy for diverse behaviors on a quadrupedal robot using scalable motion imitation," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 2768–2775. 1

[3] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "Rma: Rapid motor adaptation for legged robots," *arXiv preprint arXiv:2107.04034*, 2021. 1, 2, 4

[4] Z. Su, X. Huang, D. Ordoñez-Apraez, Y. Li, Z. Li, Q. Liao, G. Turrisi, M. Pontil, C. Semini, Y. Wu, *et al.*, "Leveraging symmetry in rl-based legged locomotion control," *arXiv preprint arXiv:2403.17320*, 2024. 1, 2

[5] G. Ji, J. Mun, H. Kim, and J. Hwangbo, "Concurrent training of a control policy and a state estimator for dynamic and robust legged locomotion," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4630–4637, 2022. 1

[6] J. Long, J. Ren, M. Shi, Z. Wang, T. Huang, P. Luo, and J. Pang, "Learning humanoid locomotion with perceptive internal model," *arXiv preprint arXiv:2411.14386*, 2024. 1

[7] Z. Zhuang, Z. Fu, J. Wang, C. Atkeson, S. Schwertfeger, C. Finn, and H. Zhao, "Robot parkour learning," *arXiv preprint arXiv:2309.05665*, 2023. 1, 2, 8

[8] J. Wu, G. Xin, C. Qi, and Y. Xue, "Learning robust and agile legged locomotion using adversarial motion priors," *IEEE Robotics and Automation Letters*, 2023. 1, 2

[9] S. Luo, S. Li, R. Yu, Z. Wang, J. Wu, and Q. Zhu, "Pie: Parkour with implicit-explicit learning framework for legged robots," *arXiv preprint arXiv:2408.13740*, 2024. 1, 2

[10] X. Cheng, K. Shi, A. Agarwal, and D. Pathak, "Extreme parkour with legged robots," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 11 443–11 450. 1, 2

[11] S. Liu, Z. Chen, Y. Liu, Y. Wang, D. Yang, Z. Zhao, Z. Zhou, X. Yi, W. Li, W. Zhang, *et al.*, "Improving generalization in visual reinforcement learning via conflict-aware gradient agreement augmentation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 23 436–23 446. 1

[12] S. Zhou, W. Zhang, J. Jiang, W. Zhong, J. Gu, and W. Zhu, "On the convergence of stochastic multi-objective gradient manipulation and beyond," *Advances in Neural Information Processing Systems*, vol. 35, pp. 38 103–38 115, 2022. 1

[13] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural computation*, vol. 3, no. 1, pp. 79–87, 1991. 1, 2

[14] J. Obando-Ceron, G. Sokar, T. Willi, C. Lyle, J. Farebrother, J. Foerster, G. K. Dziugaite, D. Precup, and P. S. Castro, "Mixtures of experts unlock parameter scaling for deep rl," *arXiv preprint arXiv:2402.08609*, 2024. 1, 2

[15] K. Li, M. Cucuringu, L. Sánchez-Betancourt, and T. Willi, "Mixtures of experts for scaling up neural networks in order execution," in *Proceedings of the 5th ACM International Conference on AI in Finance*, 2024, pp. 669–676. 1

[16] O. Celik, A. Taranovic, and G. Neumann, "Acquiring diverse skills using curriculum reinforcement learning with mixture of experts," *arXiv preprint arXiv:2403.06966*, 2024. 1, 2

[17] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, *et al.*, "Isaac gym: High performance gpu-based physics simulation for robot learning," *arXiv preprint arXiv:2108.10470*, 2021. 2, 4

[18] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, and A. Garg, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3740–3747, 2023. 2

[19] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019. 2

[20] S. Zhu, R. Huang, L. Mou, and H. Zhao, "Robust robot walker: Learning agile locomotion over tiny traps," *arXiv preprint arXiv:2409.07409*, 2024. 2, 3, 4

[21] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science robotics*, vol. 5, no. 47, p. eabc5986, 2020. 2

[22] G. B. Margolis, G. Yang, K. Paigwar, T. Chen, and P. Agrawal, "Rapid locomotion via reinforcement learning," *The International Journal of Robotics Research*, vol. 43, no. 4, pp. 572–587, 2024. 2

[23] T. He, C. Zhang, W. Xiao, G. He, C. Liu, and G. Shi, "Agile but safe: Learning collision-free high-speed legged locomotion," *arXiv preprint arXiv:2401.17583*, 2024. 2

[24] Y. Li, J. Li, W. Fu, and Y. Wu, "Learning agile bipedal motions on a quadrupedal robot," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 9735–9742. 2

[25] L. Smith, J. C. Kew, T. Li, L. Luu, X. B. Peng, S. Ha, J. Tan, and S. Levine, "Learning and adapting agile locomotion skills by transferring experience," *arXiv preprint arXiv:2304.09834*, 2023. 2

[26] Y. Cheng, H. Liu, G. Pan, L. Ye, H. Liu, and B. Liang, "Quadruped robot traversing 3d complex environments with limited perception," *arXiv preprint arXiv:2404.18225*, 2024. 2

[27] D. Hoeller, N. Rudin, D. Sako, and M. Hutter, "Anymal parkour: Learning agile navigation for quadrupedal robots," *Science Robotics*, vol. 9, no. 88, p. eadi7566, 2024. 2

[28] R. Caruana, "Multitask learning," *Machine learning*, vol. 28, pp. 41–75, 1997. 2

[29] R. Collobert and J. Weston, "A unified architecture for natural language processing: Deep neural networks with multitask learning," in *Proceedings of the 25th international conference on Machine learning*, 2008, pp. 160–167. 2

[30] S. Vandenhende, S. Georgoulis, W. Van Gansbeke, M. Proesmans, D. Dai, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 7, pp. 3614–3633, 2021. 2

[31] P. Liu, X. Qiu, and X. Huang, "Recurrent neural network for text classification with multi-task learning," *arXiv preprint arXiv:1605.05101*, 2016. 2

[32] L. Pinto and A. Gupta, "Learning to push by grasping: Using multiple tasks for effective learning," in *2017 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2017, pp. 2161–2168. 2

[33] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, "Gradient surgery for multi-task learning," *Advances in Neural Information Processing Systems*, vol. 33, pp. 5824–5836, 2020. 2

[34] B. Liu, X. Liu, X. Jin, P. Stone, and Q. Liu, "Conflict-averse gradient descent for multi-task learning," *Advances in Neural Information Processing Systems*, vol. 34, pp. 18 878–18 890, 2021. 2

[35] S. Huang, Z. Zhang, T. Liang, Y. Xu, Z. Kou, C. Lu, G. Xu, Z. Xue, and H. Xu, "Mentor: Mixture-of-experts network with task-oriented perturbation for visual reinforcement learning," *arXiv preprint arXiv:2410.14972*, 2024. 2

[36] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *International conference on machine learning*. PMLR, 2017, pp. 1126–1135. 2

[37] Y. Duan, J. Schulman, X. Chen, P. L. Bartlett, I. Sutskever, and P. Abbeel, "Rl^2: Fast reinforcement learning via slow reinforcement learning," *arXiv preprint arXiv:1611.02779*, 2016. 2

[38] S. Sodhani, A. Zhang, and J. Pineau, "Multi-task reinforcement learning with context-based representations," in *International Conference on Machine Learning*. PMLR, 2021, pp. 9767–9779. 2

E. 训练细节

为了防止腿部拖地，并在不平坦地形上实现稳健性能，我们对地面施加分形噪声 [7] 。我们使用最大噪声尺度z_{max} =0。1。我们使用PPO作为我们的强化学习算法。超参数如表IX所示表IX。

表IX：PPO超参数

超参数	数值
裁剪最小标准差	0.05
裁剪参数	0.2
折扣因子	0.99
lam	0.95
期望KL散度	0.01
熵系数	0.01
学习率	0.001
最大梯度范数	1
小批量数量	4
每个环境步数	24

参考文献

[1] K. N. Kumar、I. Essa和S. Ha，“面向复杂交互行为的级联组合残差学习”，IEEE机器人与自动化快报，第 8, 8, 4601 4608, 2023 1, 2 卷，第–页。[2] A. Klipfel、N. Sontakke、R. Liu和S. Ha，“利用可扩展运动模仿学习四足机器人多样化行为的单一策略”，载于2023年IEEE/RSJ智能机器人与系统国际会议（IROS）。IEEE, 2023, 2768 2775 1，第–页。[3] A. Kumar、Z. Fu、D. Pathak和J. Malik，“RMA：腿式机器人的快速运动适应”，arXiv预印本arXiv: *2107 04034*, 2021 1, 2, 4。. [4] Z. Su、X. Huang、D. Ordonez-Apraez、Y. Li、Z. Li、Q. Liao、G. Turrisi、M. Pontil、C. Semini、Y. Wu等人，“在基于强化学习的腿式运动控制中利用对称性”，arXiv预印本arXiv:2403.17320，2024年。1, 2[5] G. Ji、J. Mun、H. Kim和J. Hwangbo，“动态且稳健的腿式运动中控制策略与状态估计器的并发训练”，IEEE机器人与自动化快报，第7卷，4637，2022年2, 4630 1，第–页。[6] J. Long、J. Ren、M. Shi、Z. Wang、T. Huang、P. Luo和J. Pang，“利用感知内部模型学习人形机器人运动”，arXiv预印本arXiv: *2411 14386*, 2024 1。. [7] Z. Zhuang、Z. Fu、J. Wang、C. Atkeson、S. Schwertfeger、C. Finn和H. Zhao，1, 2, 8“机器人跑酷学习”，arXiv预印本arXiv:2309 05665，2023年。. [8] J. Wu、G. Xin、C. Qi和Y. Xue，“利用对抗性运动先验学习稳健且敏捷的腿式运动”，IEEE机器人与自动化快报，2023 1, 2。[9] S. Luo、S. Li、R. Yu、Z. Wang、J. Wu和Q. Zhu，“PIE：基于隐式-显式学习框架的腿式机器人跑酷”，arXiv预印本arXiv: *2408 13740*, 2024 1, 2。. [10] X. Cheng、K. Shi、A. Agarwal和D. Pathak，“腿式机器人极限跑酷”，载于2024年IEEE机器人与自动化国际会议（ICRA）。IEEE, 2024, 11 443 11 450 1, 2，第–页。[11] S. Liu、Z. Chen、Y. Liu、Y. Wang、D. Yang、Z. Zhao、Z. Zhou、X. Yi、W. Li、W. Zhang等人，“通过冲突感知梯度一致性增强改进视觉强化学习中的泛化性”，载于IEEE/CVF国际计算机视觉会议论文集，2023, 23 436 23 446 1，第–页。[12] S. Zhou、W. Zhang、J. Jiang、W. Zhong、J. Gu和W. Zhu，“关于随机多目标梯度操纵的收敛性及其扩展”，神经信息处理系统进展，第35卷，第 38 103 38 115, 2022 1 –页。[13] R. A. Jacobs、M. I. Jordan、S. J. Nowlan和G. E. Hinton，“自适应局部专家混合”，神经计算，第3卷，87，1991年1, 79, 1, 2 第–页。[14] J. Obando-Ceron、G. Sokar、T. Willi、C. Lyle、J. Farebrother、J. Foerster、G. K. Dziugaite、D. Precup和P. S. Castro，“专家混合解锁深度强化学习的参数扩展”，arXiv预印本arXiv:2402.08609，2024年。1, 2 [15] K. Li、M. Cucuringu、L. S´anchez-Betancourt和T. Willi，“用于订单执行中神经网络扩展的专家混合”，载于第5届ACM人工智能金融国际会议论文集，2024年，第669–676页。1

[16] O. Celik、A. Taranovic和G. Neumann，“利用专家混合的课程强化学习获取多样化技能”，arXiv预印本，arXiv: *2403 06966*, 2024 1, 2。. [17] V. Makoviychuk、L. Wawrzyniak、Y. Guo、M. Lu、K. Storey、M. Macklin、D. Hoeller、N. Rudin、A. Allshire、A. Handa等人，“Isaac Gym：用于机器人学习的高性能基于GPU的物理仿真”，arXiv预印本，arXiv: *2108 10470*, 2021 2, 4。. [18] M. Mittal、C. Yu、Q. Yu、J. Liu、N. Rudin、D. Hoeller、J. L. Yuan、R. Singh、Y. Guo、H. Mazhar、A. Mandlekar、B. Babich、G. State、M. Hutter和A. Garg，“Orbit：用于交互式机器人学习环境的统一仿真框架”，IEEE机器人与自动化快报，第8卷，第 6, 3740 3747, 2023 2期，第–页。. [19] J. Hwangbo、J. Lee、A. Dosovitskiy、D. Bellicoso、V. Tsounis、V. Koltun和M. Hutter，“学习腿式机器人的敏捷动态运动技能”，科学机器人学，第 4, 26, 5872, 2019 2卷，第期，第eaau [20] 页。. S. Zhu、R. Huang、L. Mou和H. Zhao，“鲁棒机器人步行器：在微小陷阱上学习敏捷运动”，arXiv预印本，arXiv:2409 07409，2024年，第2、3、4页。. . [21] J. Lee、J. Hwangbo、L. Wellhausen、V. Koltun 和 M. Hutter，“在具有挑战性的地形上学习四足运动”，科学机器人学，第 5, 47, 5986, 2020 2 卷，第期，第eabc页。. [22] G. B. Margolis、G. Yang、K. Paigwar、T. Chen和P. Agrawal，“通过强化学习实现快速运动”，国际机器人研究杂志，第 43, 4, 572 587, 2024 2 卷，第期，第–页。. [23] T. He、C. Zhang、W. Xiao、G. He、C. Liu和G. Shi，“敏捷但安全：学习无碰撞高速腿式运动”，arXiv预印本，arXiv:2401.17583，2024年。. 2[24] Y. Li、J. Li、W. Fu和Y. Wu，“在四足机器人上学习敏捷双足运动”，载于2024年IEEE机器人与自动化国际会议（ICRA）。IEEE, 2024, 9735 9742 2年，第–页。. [25] L. Smith、J. C. Kew、T. Li、L. Luu、X. B. Peng、S. Ha、J. Tan和S. Levine，“通过经验迁移学习并适应敏捷运动技能”，arXiv预印本，arXiv: *2304 09834*, 2023 2。. [26] Y. Cheng、H. Liu、G. Pan、L. Ye、H. Liu和B. Liang，“四足机器人在有限感知下穿越三维复杂环境”，arXiv预印本，arXiv: *2404 18225*, 2024 2 。. [27] D. Hoeller、N. Rudin、D. Sako和M. Hutter，“Anymal跑酷：学习四足机器人的敏捷导航”，科学机器人学，第 9, 88, 7566, 2024 2卷，第期，第eadi页。. [28] R. Caruana，“多任务学习”，机器学习，第28卷，第75页，1997年，第41、2页，第–页。. [29] R. Collobert和J. Weston，“自然语言处理的统一架构：基于多任务学习的深度神经网络”，载于第25届国际机器学习会议论文集，2008年，第 160 167 2 –页。. [30] S. Vandenhende、S. Georgoulis、W. Van Gansbeke、M. Proesmans、D. Dai和L. Van Gool，“密集预测任务的多任务学习：综述”，IEEE模式分析与机器智能汇刊，第 44, 7, 3614 3633, 2021 2 卷，第期，第–页。. [31] P. Liu、X. Qiu和X. Huang，“基于多任务学习的文本分类循环神经网络”，arXiv预印本，arXiv:1605 05101，2016年，第2页。. . [32] L. Pinto和A. Gupta，“通过抓取学习推动：利用多任务进行有效学习”，载于2017年IEEE国际机器人与自动化会议（ICRA）。IEEE, 2017, 2161 2168 2年，第–页。. [33] T. Yu、S. Kumar、A. Gupta、S. Levine、K. Hausman和C. Finn，“多任务学习的梯度手术”，神经信息处理系统进展，第 33, 5824 5836, 2020 2卷，第–页。. [34] B. Liu、X. Liu、X. Jin、P. Stone和Q. Liu，“多任务学习的冲突规避梯度下降”，神经信息处理系统进展，第 34, 18 878 18 890, 2021 2卷，第–页。. [35] S. Huang、Z. Zhang、T. Liang、Y. Xu、Z. Kou、C. Lu、G. Xu、Z. Xue和H. Xu，“Mentor：面向视觉强化学习的任务导向扰动专家混合网络”，arXiv预印本，arXiv:2410.14972，2024年。. 2[36] C. Finn、P. Abbeel和S. Levine，“用于深度网络快速适应的模型无关元学习”，载于国际机器学习会议。PMLR, 2017, 1126 1135 2年，第–页。. [37] Y. Duan、J. Schulman、X. Chen、P. L. Bartlett、I. Sutskever和P. Abbeel，“Rl^2：通过慢速强化学习实现快速强化学习”，arXiv预印本，arXiv: *1611 02779*, 2016 2。. [38] S. Sodhani、A. Zhang和J. Pineau，“基于上下文的表示的多任务强化学习”，载于国际机器学习会议。PMLR，2021年，第9767–9779页。. 2

[39] C. Yang, K. Yuan, Q. Zhu, W. Yu, and Z. Li, “Multi-expert learning of adaptive legged locomotion,” *Science Robotics*, vol. 5, no. 49, p. eabb2174, 2020. 2

[40] Y. Ze, G. Yan, Y.-H. Wu, A. Macaluso, Y. Ge, J. Ye, N. Hansen, L. E. Li, and X. Wang, “Gnfactor: Multi-task real robot learning with generalizable neural feature fields,” in *Conference on Robot Learning*. PMLR, 2023, pp. 284–301. 2

[41] M. Shafiee, G. Bellegarda, and A. Ijspeert, “Manyquadrupeds: Learning a single locomotion policy for diverse quadruped robots,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 3471–3477. 2

[42] C. Yang, K. Yuan, Q. Zhu, W. Yu, and Z. Li, “Multi-expert learning of adaptive legged locomotion,” *Science Robotics*, vol. 5, no. 49, p. eabb2174, 2020. 2

[43] N. Shah, K. Tiwari, and A. Bera, “Mtac: Hierarchical reinforcement learning-based multi-gait terrain-adaptive quadruped controller,” *arXiv preprint arXiv:2401.03337*, 2023. 2

[44] M. I. Jordan and R. A. Jacobs, “Hierarchical mixtures of experts and the em algorithm,” *Neural computation*, vol. 6, no. 2, pp. 181–214, 1994. 2

[45] M. Deisenroth and J. W. Ng, “Distributed gaussian processes,” in *International conference on machine learning*. PMLR, 2015, pp. 1481–1490. 2

[46] D. Lepikhin, H. Lee, Y. Xu, D. Chen, O. Firat, Y. Huang, M. Krikun, N. Shazeer, and Z. Chen, “Gshard: Scaling giant models with conditional computation and automatic sharding,” *arXiv preprint arXiv:2006.16668*, 2020. 2

[47] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D. S. Chaplot, D. d. l. Casas, E. B. Hanna, F. Bressand, *et al.*, “Mixtral of experts,” *arXiv preprint arXiv:2401.04088*, 2024. 2

[48] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan, *et al.*, “Deepseek-v3 technical report,” *arXiv preprint arXiv:2412.19437*, 2024. 2

[49] Z. Zhang, S. Liu, J. Yu, Q. Cai, X. Zhao, C. Zhang, Z. Liu, Q. Liu, H. Zhao, L. Hu, *et al.*, “M3oe: Multi-domain multi-task mixture-of experts recommendation framework,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 893–902. 2

[50] J. Jiang, P. Zhang, Y. Luo, C. Li, J. B. Kim, K. Zhang, S. Wang, X. Xie, and S. Kim, “Adamct: adaptive mixture of cnn-transformer for sequential recommendation,” in *Proceedings of the 32nd ACM international conference on information and knowledge management*, 2023, pp. 976–986. 2

[51] Y. Gou, Z. Liu, K. Chen, L. Hong, H. Xu, A. Li, D.-Y. Yeung, J. T. Kwok, and Y. Zhang, “Mixture of cluster-conditional lora experts for vision-language instruction tuning,” *arXiv preprint arXiv:2312.12379*, 2023. 2

[52] S. Chen, Z. Jie, and L. Ma, “Llava-mole: Sparse mixture of lora experts for mitigating data conflicts in instruction finetuning mllms,” *arXiv preprint arXiv:2401.16160*, 2024. 2

[53] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017. 3

[54] S. Zhu, D. Li, L. Mou, Y. Liu, N. Xu, and H. Zhao, “Saro: Space-aware robot system for terrain crossing via vision-language model,” *arXiv preprint arXiv:2407.16412*, 2024. 3

[39] C. Yang, K. Yuan, Q. Zhu, W. Yu, and Z. Li,“自适应腿式运动的多专家学习，”科学机器人学, 第5卷，第49期，第eabb2174页，2020年。2[40] Y. Ze, G. Yan, Y.-H. Wu, A. Macaluso, Y. Ge, J. Ye, N. Hansen, L. E. Li, and X. Wang,“Gnfactor：基于可泛化的神经特征场的多任务真实机器人学习，”载于机器人学习会议. PMLR, 2023, 284 301 2 第–页。[41] M. Shafiee, G. Bellegarda, and A. Ijspeert,“ManyQuadrupeds：为多样化四足机器人学习单一运动策略，”载于*2024年IEEE*机器人与自动化国际会议（*ICRA*）. IEEE, 2024, 3471 3477 2 第–页。C. Yang, K. Yuan, [42] Q. Zhu, W. Yu, and Z. Li,“自适应腿式运动的多专家学习，”科学机器人学，第 5 卷，第 49 期，第 eabb2174 页，2020 年。2[43] N. Shah, K. Tiwari, and A. Bera,“MTAC：基于分层强化学习的多步态地形自适应四足控制器，”*arXiv*预印本 *arXiv: 2401 03337*, 2023 2 .. M. I. Jordan [44] and R. A. Jacobs,“专家分层混合与EM算法，”神经计算, 第 6, 2, 181 214, 1994 2 卷，第–[45] 页。M. Deisenroth and J. W. Ng,“分布式高斯过程，”载于国际机器学习会议. PMLR, 2015, 第1481–1490页。2[46] D. Lepikhin, H. Lee, Y. Xu, D. Chen, O. Firat, Y. Huang, M. Krikun, N. Shazeer, and Z. Chen,“Gshard：通过条件计算和自动分片扩展巨型模型，”*arXiv*预印本 *arXiv:2006.16668*, 2020年。2[47] A. Q. Jiang, A. Sablayrolles, A.Roux, A. Mensch, B.Savary, C. Bam- ford, D. S. Chaplot, D. d. l. Casas, E. B. Hanna, F. Bressand, 等,

“专家混合模型Mixtral，” arXiv预印本arXiv:2401.04088, 2024年。2[48] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan, 等，“DeepSeek-V3技术报告，” arXiv 预印本 arXiv:2412.19437，2024 年。2[49] Z. Zhang, S. Liu, J. Yu, Q. Cai, X. Zhao, C. Zhang, Z. Liu, Q. Liu, H. Zhao, L. Hu, 等，“M3oe：多域多任务专家混合推荐框架，” 第47届国际ACM SIGIR信息检索研究与发展会议论文集，2024年，第893–902页。2

[50] J. Jiang, P. Zhang, Y. Luo, C. Li, J. B. Kim, K. Zhang, S. Wang, X. Xie, 和 S. Kim，“Adamct：用于序列推荐的自适应CNN-Transformer混合，” 第32届ACM国际信息与知识管理会议论文集，2023年，第976–986页。2[51] Y. Gou, Z. Liu, K. Chen, L. Hong, H. Xu, A. Li, D.-Y. Yeung, J. T. Kwok, 和 Y. Zhang，“用于视觉-语言指令微调的簇条件LoRA专家混合，” arXiv预印本arXiv:2312 12379，2023年2。[52] S. Chen, Z. Jie, 和 L. Ma，“Llava-mole：用于缓解多模态大语言模型指令微调中数据冲突的稀疏LoRA专家混合，” arXiv预印本arXiv:2401.16160，2024年。2[53] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, 和 O. Klimov,“近端策略优化算法，” arXiv预印本arXiv:1707 06347，2017年3。

[54] S. Zhu, D. Li, L. Mou, Y. Liu, N. Xu, 和 H. Zhao，“Saro：基于视觉-语言模型的地形穿越空间感知机器人系统，” arXiv预印本arXiv:2407.16412，2024年。3